

TRI-NETRA ADCS: design loop, SILS, Monte Carlo and soft OILS

From each customer case to a sized, selected and dispatched attitude-control configuration, verified on the Rust engine (POP v51 in the loop), against the MATLAB twin, and with the flight software running as Cortex-M4F firmware.

AIS_3U SELECTED mtq_fmr feasible	AIS_IMG_3U SELECTED mtq_fmr feasible	ENGINE VS MATLAB VERDICTS 94/107 truth environment identical
SILS VS SOFT OILS VERDICTS 105/107 worst OBC load 6.8 %	VIRTUAL OBC BIT-IDENTICAL 35 runs, process + QEMU	CAMPAIGNS ON THE ENGINE 8 MC + edge, vs MATLAB

Owner	Agastya
Generated	2026-09-30 16:57 UTC
Repository revision	86490e6
Engine	adcs-engine-rs/1.2.0 (adcs-case/1, POP v51 port in-loop, adcs-design, soft OILS)
Flight software	trinetra-fsw-c/1.0.0 and trinetra-fsw-rs (C99 and Rust no_std, adcs-fswcfg/1)
Cases	ais_3u, ais_img_3u
Regenerate	<code>python3 tools/pipeline.py · python3 tools/engine.py campaign · python3 tools/engine.py oils · python3 tools/vv_report.py</code>

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1. Summary and findings

ais_3u (coarse class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), feasible, after 6 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.446 kg, 0.537 L. Best benchmark (catalogue actuators, same rule): `mtq_rw`, closest (not feasible), 0.393 kg — `sun_referencing: power_mean` (power).

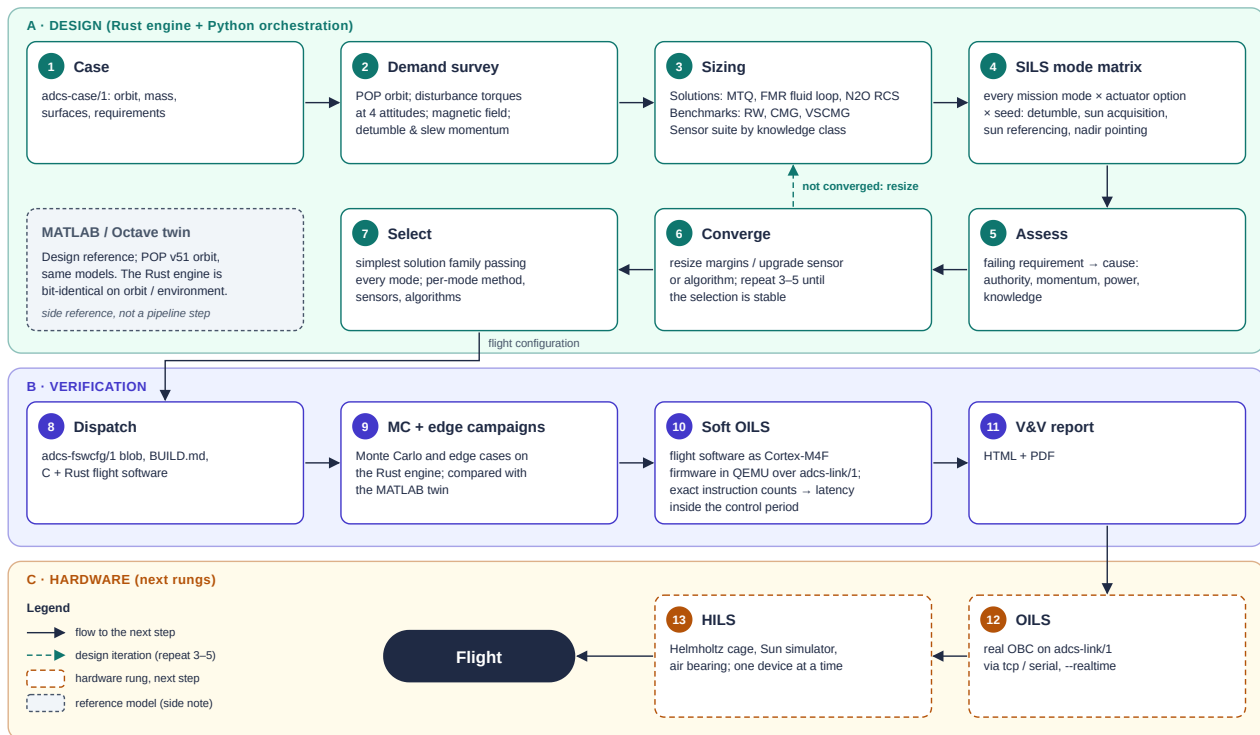
ais_img_3u (fine class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), feasible, after 6 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.427 kg, 0.534 L. Best benchmark (catalogue actuators, same rule): `mtq_rw`, feasible, 0.696 kg.

2. Architecture and flow

The design loop sizes every actuator option to the case, flies every mission mode with every option in the SILS, converges the sizing, sensors and algorithms, selects the lightest of our solution families that passes, dispatches its flight configuration and verifies it by Monte Carlo and in soft OILS. The same flight software (C99 and Rust, one pseudocode contract) runs in every rung behind one byte-level HAL.

TRI-NETRA ADCS: from customer case to flight

End-to-end V&V flow: design loop, software verification, then hardware-in-the-loop rungs



TRI-NETRAADCS - Agastya

Figure 1. From customer case to flight: design loop, verification, hardware rungs.

Who runs what: languages and layers

TRI-NETRA ADCS, from the flight contract at the top to the ground tooling and outputs at the bottom

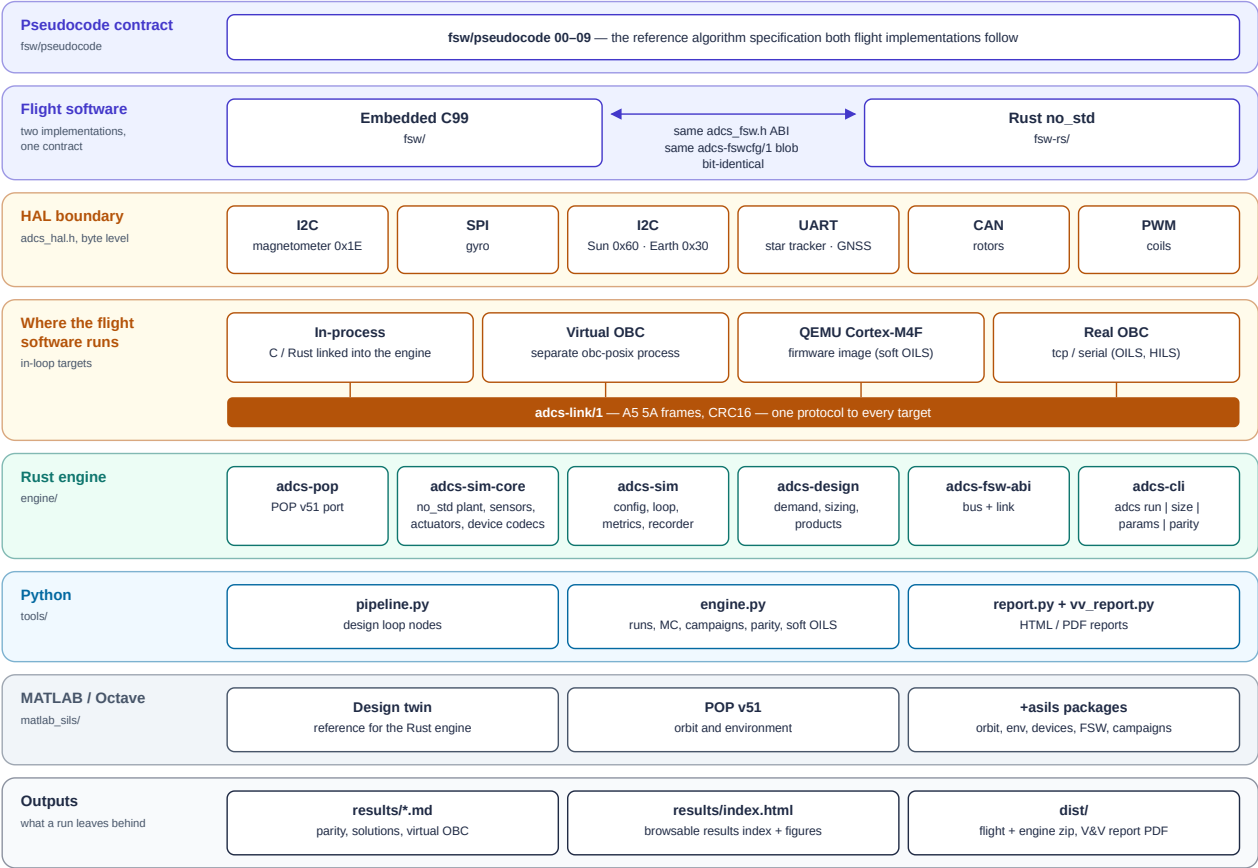


Figure 2. Languages and layers.

3. Nodes

Every step of the chain is a node with its inputs, outputs, parameters and rules in one registry (`matlab_sils/data/pipeline/nodes.json`; `docs/NODES.md`). The pipeline reads its parameters from it, so this table is what ran.

#	node	stage	runs in	what it does	parameters	rules
1	case	input	file	The customer case: orbit, mass properties, surfaces, requirements and budgets (adcs-case/1).	—	every requirement source (stated by UNCONFIRMED reference) the case file is pre-cached run's key requirement re-fl
2	catalogue	input	file + Python (tools/catalogue.py derives the modelling block)	Bought actuators as their datasheets state them: CubeSpace, AAC Clyde Space and Rocket Lab reaction wheels, Tensor Tech control moment gyroscopes. One file per model with the datasheet numbers, the source URL and the model parameters derived from them (each derived value says how).	vendors = CubeSpace, AAC Clyde Space, Rocket Lab, Tensor Tech	a number not on is null in the data never guessed model parameter needs but the data (rotor inertia, friction imbalance) are derived, RW, CMG and V bought: they are the catalogue, not a law

#	node	stage	runs in	what it does	parameters	rules
3	demand	design	Rust (adcs-design::demand)	What the case asks of an actuator: one orbit on the POP orbit at four held attitudes with the SILS torque models; peak disturbance, cyclic and secular momentum, weakest field, detumble and slew momentum.	k_h = momentum margin: 2, or 1/(1 - req.hsat) k_tau = 1.5	h_req = k_h x m secular per orbit, momentum) tau_req = k_tau : disturbance, slew fine class when r 0.05 deg (star tra precision gyro)
4	size_mtg	design	Rust (adcs-design::mtg)	Our magnetorquer coils, sized: the largest of the dumping, momentum and detumble dipoles.	floor_Am2 = 0.05 anchor = SYN-CT-1: mass 0.03 kg, 0.3 W per 0.45 Am2	m = max(2 tau_d B_min), 2 h_sec B_mean T), h_d B_mean t_deturr the coils-only fan the pointing-grad least twice the d dipole)
5	size_fmr	design	Rust (adcs-design::fmr, empump)	Our fluid momentum loop, one ring per axis, designed with its electromagnetic DC conduction pump (no permanent magnet).	electrode_current_max_A = 10 iron_flux_T = 1.2 fluid = galinstan cruise_fraction = 0.4	grid over flow sp gap field and pur keep the least m x steady power tl h_req and tau_re torque capacity = x 2 S A / L; coil r sqrt(lambda K)
6	size_rcs	design	Rust (adcs-design::rcs)	Our N2O cold-gas RCS, 6 couples: thrust class and propellant tank.	isp_budget_s = 60 thrust_classes_N = 0.005, 0.01, 0.02, 0.05, 0.1 meop_bar = 70	thrust: the smalle above the couple detumble and sle propellant: detur dumping over life

#	node	stage	runs in	what it does	parameters	rules
7	<code>select_rotor</code>	design	Rust (adcs-design::catalogue)	The benchmarks' momentum actuators chosen from the catalogue: for RW three orthogonal wheels, for CMG and VSCMG a four-unit 54.74 deg pyramid.	rw_units = 3 cmg_units = 4 pyramid_deg = 54.74 per_unit_need = rw: h_max >= h_req x scale and torque >= tau_req x scale (three wheels, one per axis); cmg: h_max >= h_req/2 x scale and torque >= tau_req/2 x scale (a pyramid puts about two units on any axis); vscmg: as cmg; the rotor runs at half its momentum so its speed can move both ways choose = the lightest model that meets the need; ties by steady power, then volume selectable = a catalogue model whose datasheet states momentum, torque, mass and steady power (tools/catalogue.py)	when no model r need the largest the gap is reporte cannot size it up) an authority incre converge node r need, so the nex chosen
8	<code>size_sensors</code>	design	Rust (adcs-design::size_all)	The sensor set of every product: magnetometer, six sun sensors, gyro, GNSS, Earth sensor; the star tracker on the fine class or when knowledge asks for it.	star_tracker = SYN-ST-1 st_heads_default = 2 gyro_fine = TRN-GYRO-P1 gyro_coarse = SYN-GYRO-1	gyro grade g: noi and power / g (fit below 1)
9	<code>budget</code>	design	Rust (adcs-design::budget)	Each family's product: mass, steady power and volume summed over every fitted unit.	—	units per fill = nu spin axes / bores sun sensors cou

#	node	stage	runs in	what it does	parameters	rules
10	matrix	SILS	Rust engine + C flight software (adcs run)	Every mission mode x actuator option x seed on the sized products; options that fail on performance also fly every algorithm of their slot.	seeds = 1, 2 cache_key = scenario, product, parts, seed, flight-software build, case file algorithm_candidates = detumble: bdot_gyro, bdot_mag, bdot_bangbang, genbdot_l1; sun_acquisition: sunspin_damped, sunspin_l1l2, sunspin_l1l2_e2, sunspin_deruiter2011, sun_boresight_celani2026; mtq_pointing: mtq_pd, mtq_lqr, mtq_smc, mtq_rate_damp, mtq_lovera2004, mtq_celani2015, mtq_avanzini2021, mtq_celani2026, mtq_tango2013; pointing: pid, lqr, smc, pid@bw2.5, pid@bw4	a run is scored a case's requirement time channels are del scoring (the man the coils-only law magnetorquer lite (docs/MTQ_LITE are candidates o like every other l
11	assess	SILS	Python (tools/pipeline.py)	Per option: feasible on every seed? Each failing requirement classed; the best algorithm per option kept.	classes = power*: power; ake*: knowledge; propellant*: propellant; otherwise: performance violation = value / requirement - 1 (10 when not finite)	rank an option's (not feasible, faili worst objective)

#	node	stage	runs in	what it does	parameters	rules
12	tune	SILS	Rust engine + C flight software (through matrix)	Bruni & Celani 2017 (P7) min-max gain selection: for a coils-only option that still fails on performance after every law of its slot has flown, or passes with a thin margin,, every law is flown at every point of a gain grid, on extra seeds; each law keeps the gains whose worst seed is best, and assess then ranks the laws by that worst case.	grids = mtq_pointing: mtq_gain_p: 0.25, 1, 4; mtq_gain_d: 0.25, 1, 4; handover_out_dps: 0.25, 0.5, 1.0; sun_acquisition: spin_rate_dps: 2, 4, 6; ss_gain: 0.3, 1, 3 extra_seeds = 3, 4 actuators = mtq objective = worst seed: (not feasible, failing count, objective) margin = 0.5	mtq_gain_p / mtr scale the proport gains of every m pointing law (for lambda and k) spin_rate_dps se commanded spir wobble grows wi ss_gain the Sun- the grid is a deriv search as in the (3 x 3) to keep th inside minutes handover_out_d error below whicl law takes over fr after the Sun spi (05_control.md); with the gains be coils-only nadir to the spin an option that pa whose worst see than margin x its is tuned too (the finds thin margin:
13	converge	design	Python (tools/pipeline.py)	The knob changes the failures call for; converged when nothing is left to change.	scale_bounds = 0.5, 4.0 scale_up = 1.5 scale_down = 0.75 improvement_needed = 0.05 fmr_lambda_bounds_kg_W = 0.01, 3.0 fmr_flow_sigma_min_m_s = 5e-05 gyro_grade_min = 0.1 max_iterations = 24	performance: fly algorithm of the s (fluid loop) a quick sensor /4, then a undone when the does not fall by 5 rate stability (fine noise x0.3 down undone when it c improve knowledge: fit the on a coarse prod power, fluid loop: lambda x3 (not for power); coils-only authority x0.75; r blocked (standby floor) mass over budget solution family: li one star-tracker l /3, fluid-loop mor -- each undone a breaks a mode o violation by more propellant: RCS : a part pushed up failure and down frozen and report conflict

#	node	stage	runs in	what it does	parameters	rules
14	<code>select</code>	selection	Python (tools/pipeline.py)	Scores every family (ours and the benchmarks) on the same modes and budget, then picks the minimum feasible solution; the benchmarks are ranked by the same rule as the comparison.	feasible = every mode passes with a usable option on every seed, and $\text{mass} \leq \text{req.mass}$, $\text{volume} \leq \text{req.vol}$ rank_feasible = mass_kg, power_W, volume_L, simplicity rank_infeasible = gap_count, mass_kg select_role = solution compare_role = benchmark	selected = the feasible solution family for rank_feasible or mass, then power volume) when no solution feasible the closest named with its gap benchmark = the ranking over the families, reported selection
15	<code>dispatch</code>	dispatch	Rust (adcs dispatch)	The selected family's mission (detumble -> Sun acquisition -> nadir), its adcs-fswcfg/1 blob, the converged products; C and Rust checked bit-identical.	nadir_at_orbits = 2 nadir_orbits = rotors: 1; coils_only: 3 duration_orbits = nadir_at_orbits + nadir_orbits: detumble and Sun acquisition, then nadir (the metrics take the last half orbit)	the C and Rust files must agree bit for a coils-only nadir three orbits after command: the Sun over (despin, the capture) takes at
16	<code>certify</code>	verification	Python (tools/floquet.py)	Celani 2026 (P8) method: Floquet multipliers of the coils-only nadir loop linearised about the reference, with the gains the coils-only family dispatches and the magnetic field along one orbit; all $ \mu < 1$ certifies local exponential stability of the periodic loop.	field = aligned dipole, $7.94 \times 10^{-5} \text{ T m}^3$, in the orbit frame frame = reference body frame from the mission's own nadir phase (truth attitude) steps_per_orbit = 2000	$\theta_{\dot{}} = \omega$, $J \dot{\Gamma}(t) \tau(\theta)$ $\Gamma = I - b b^T$ and gravity-gradient out, as in the paper averaging analysis a boresight law for rotation about its free: one multiplier by design

#	node	stage	runs in	what it does	parameters	rules
17	mc	verification	Rust engine	Monte Carlo of the dispatched mission with the case's dispersions (truth dispersed, flight software nominal). It feeds back: a requirement that fails in any dispersed run is a failure the loop must fix.	runs = 12 robustness_passes = 3	dispersions as asils.campaign.d every requirement pass in every run for the selected f power -> pump v copper (lambda > lighter-pump level performance -> t authority x1.5 an authority-down level knowledge -> the on a coarse prod the loop then run new knobs (sizing converge, select, mc) up to robust times
18	family_missions	verification	Rust engine + C and Rust flight software	Every one of our solution families, selected or not (coils only, coils + fluid loop, coils + fluid loop + RCS), dispatched and flown as the full mission (detumble -> Sun acquisition -> nadir) with its best methods from the loop, in C and Rust, and by Monte Carlo, so each family's behaviour is on record beside the selection.	role = solution monte_carlo_runs = as node mc	the selected family dispatch and mc a family that is not still flown: its gap mission metrics C and Rust must bit for every family
19	soft_oils	verification	Rust engine + QEMU (Cortex-M4F)	The dispatched mission with the flight software as firmware (C and Rust) beside its SILS run; exact instruction timing and bus latency.	obc_mhz = 168 cpi = 1.25 i2c_khz = 400 spi_mhz = 1 can_kbps = 1000	a latency >= the counts as an overflow a missing instruction stops the run

#	node	stage	runs in	what it does	parameters	rules
20	report	report	Python	The design ledger per case and the V&V report (HTML and PDF).	—	every table is generated from the stored node content and typed

4. Actuator catalogue (datasheets)

Reaction wheels, CMGs and VSCMGs are bought, so the benchmarks fly real products: the `select_rotor` node takes the lightest catalogue model that meets the case's per-unit momentum and torque (three wheels; a four-unit CMG pyramid, two units per axis). We design only the magnetorquers, the fluid loop and the RCS. Numbers are the vendors'; a model whose datasheet lacks momentum, torque, mass or steady power is listed but not selectable. Values the engine needs and datasheets omit (rotor inertia, friction, imbalance) are derived with the rules in each file (`tools/catalogue.py`, `docs/CATALOGUE.md`).

vendor	model	type	h [mNms]	torque [mNm]	mass [g]	steady [W]	peak [W]	size [mm]	selec
AAC Clyde Space	RW222-3	reaction_wheel	3	2	—	—	—	25 x 25 x 15	no: m power
AAC Clyde Space	RW222-6	reaction_wheel	6	2	—	—	—	25 x 25 x 15	no: m power
AAC Clyde Space	RW400-15	reaction_wheel	15	—	—	—	15	50 x 50 x 27	no: to mass_ power
AAC Clyde Space	RW400-30	reaction_wheel	30	—	—	—	15	50 x 50 x 27	no: to mass_ power
AAC Clyde Space	RW400-50	reaction_wheel	50	—	375	—	15	50 x 50 x 27	no: to power
CubeSpace	CubeWheel CW0017 (Gen2)	reaction_wheel	1.77	0.23	60	0.3	0.85	28 x 28 x 26	yes

vendor	model	type	h [mNms]	torque [mNm]	mass [g]	steady [W]	peak [W]	size [mm]	selec
CubeSpace	CubeWheel CW0057 (Gen2)	reaction_wheel	5.7	2	115	0.77	2.7	35 x 35 x 24	yes
CubeSpace	CubeWheel CW0162 (Gen2)	reaction_wheel	16.2	7	144	0.77	7.2	46 x 46 x 24	yes
CubeSpace	CubeWheel CW0500 (Gen2)	reaction_wheel	50	16	310	5.3	15	66 x 66 x 25	yes
CubeSpace	CubeWheel CW1200 (Gen2)	reaction_wheel	120	20	450	4.7	13	72 x 72 x 31	yes
CubeSpace	CubeWheel CW2500 (Gen2)	reaction_wheel	250	27	750	9	30	86 x 86 x 36	yes
CubeSpace	CubeWheel CW5000 (Gen2)	reaction_wheel	500	37	970	9	48	100 x 100 x 37	yes
Rocket Lab	RW3-0.06	reaction_wheel	60	20	226	0.5	23.4	77 x 65 x 38	yes
Rocket Lab	RW3-1.0	reaction_wheel	1000	50	970	2.4	46.7	150 x 140 x 42	yes
Rocket Lab	RW-0.003	reaction_wheel	3	1	50	—	—	33.5 x 33.5 x 17	no: power
Rocket Lab	RW-0.01	reaction_wheel	10	1	120	0.11	1.05	50 x 50 x 30	yes

vendor	model	type	h [mNms]	torque [mNm]	mass [g]	steady [W]	peak [W]	size [mm]	selec
Rocket Lab	RW-0.03	reaction_wheel	30	2	183	0.1	1.8	50 x 50 x 40	yes
Tensor Tech	ADCS400 (4 x RS100 variable- speed CMG pyramid)	cmg_cluster	10	1	300	1	1	0.2U+ tuna can per unit, >64 mm dia	yes
Tensor Tech	TensorCMG- 10m	cmg	13	100	—	—	—	—	no: m power

The vendor sites could not be reached from the build environment; numbers marked verified were read from the full datasheet PDFs (distributor mirror), the others from search excerpts of the cited page and should be confirmed with the vendor. Tensor Tech's ADCS400 figures are the whole integrated ADCS (CMGs, magnetorquer, sun sensors, gyro) as upper bounds, so the CMG benchmark's mass and power are conservative.

5. Requirements

ais_3u — 3U AIS satellite — coarse pointing 10 deg, SSO dawn-dusk 550 km

key	quantity	value	unit	source / status
req.ape	Absolute pointing error (APE)	10.0	Degree	stated by the user: the AIS antenna needs only coarse nadir pointing
req.ake	Absolute knowledge error (AKE)	5.0	Degree	UNCONFIRMED · plan reference: half the APE, so knowledge never dominates the budget
req.detumble	Detumble time	284.0	Minute	UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec
req.sunacq	Sun-acquisition time in safe mode	95.0	Minute	UNCONFIRMED · plan reference: one orbit
req.mass	ADCS mass	1.6	Kilogram	set by the user (Agastya): the same ADCS budget in every 3U case, 1U of the 3U, 1.5-1.7 kg; 1.6 keeps 0.1 kg to the 1.7 ceiling
req.pavg	ADCS orbit-average power	0.5	Watt	UNCONFIRMED · plan reference: leaves the AIS receiver its share of a 3U power budget
req.ppk	ADCS peak power	1.5	Watt	UNCONFIRMED · plan reference: all coils at full current
req.vol	ADCS volume	1.0	Litre	set by the user (Agastya): 1U of the 3U is the ADCS
mission.epoch	Launch epoch	27.0	Year	UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027
mission.spd	Slews per day	10.0	Count	UNCONFIRMED · adcs_ref_c1
mission.sangle	Reference slew angle	30.0	Degree	UNCONFIRMED · adcs_ref_c1
mission.w0	Initial tumble rate after separation	10.0	DegreePerSecond	UNCONFIRMED · adcs_ref_c1: tumble after separation
orbit.alt	Altitude	550.0	Kilometre	stated by the user (Agastya): 550 km LEO
orbit.inc	Inclination	97.593	Degree	sun-synchronous at 550 km: asils.orbit.sso_inclination(550)
orbit.ecc	Eccentricity	0.0	One	UNCONFIRMED · adcs_ref_c1
orbit.ltan	LTAN	6.0	Hour	stated by the user: SSO dawn-dusk (LTAN 06:00)
mass.m	Satellite mass	4.0	Kilogram	UNCONFIRMED · adcs_ref_c1
mass.imax	Largest principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12 \cdot (0.10^2 + 0.34^2)$
mass.iint	Intermediate principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal

key	quantity	value	unit	source / status
mass.imin	Smallest principal inertia	0.0067	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.10^2)$

ais_img_3u — 3U imaging satellite — fine pointing 0.01 deg, SSO 10:00 550 km

key	quantity	value	unit	source / status
req.ape	Absolute pointing error (APE)	0.01	Degree	stated by the user; 3-sigma. A flown 3U reference: MinXSS-1 (BCT XACT) measured 0.0042-0.0117 deg 3-sigma, axis dependent (arXiv:1706.06967)
req.ake	Absolute knowledge error (AKE)	0.005	Degree	UNCONFIRMED · plan reference: half the APE
req.rks	Rate stability	0.005	DegreePerSecond	relaxed by the user (Agastya) from 0.001: 18 arcsec/s; at 550 km that is 48 m/s of ground motion, 0.5 m of smear in a 10 ms exposure
req.slew	Reference slew time	60.0	Second	UNCONFIRMED · adcs_ref_c1: mission.sangle in 60 s, target to target
req.settle	Settling time after a slew	20.0	Second	UNCONFIRMED · plan reference: back inside the APE after a slew
req.detumble	Detumble time	284.0	Minute	UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec
req.sunacq	Sun-acquisition time in safe mode	95.0	Minute	UNCONFIRMED · plan reference: one orbit
req.mass	ADCS mass	1.6	Kilogram	raised by the user (Agastya): 1.5-1.7 kg with 1U of the 3U for the ADCS; 1.6 keeps 0.1 kg to the 1.7 ceiling
req.pavg	ADCS orbit-average power	2.0	Watt	UNCONFIRMED · plan reference
req.ppk	ADCS peak power	4.0	Watt	UNCONFIRMED · plan reference
req.vol	ADCS volume	1.0	Litre	raised by the user (Agastya): 1U of the 3U is the ADCS
mission.epoch	Launch epoch	27.0	Year	UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027
mission.spd	Slews per day	10.0	Count	UNCONFIRMED · adcs_ref_c1
mission.sangle	Reference slew angle	30.0	Degree	UNCONFIRMED · adcs_ref_c1
mission.w0	Initial tumble rate after separation	10.0	DegreePerSecond	UNCONFIRMED · adcs_ref_c1: tumble after separation
orbit.alt	Altitude	550.0	Kilometre	stated by the user (Agastya): 550 km LEO
orbit.inc	Inclination	97.593	Degree	sun-synchronous at 550 km: asils.orbit.sso_inclination(550)
orbit.ecc	Eccentricity	0.0	One	UNCONFIRMED · adcs_ref_c1

key	quantity	value	unit	source / status
orbit.ltan	LTAN	10.0	Hour	stated by the user: SSO 10:00 (read as LTAN 10:00; LTDN 22:00 gives the same plane and beta magnitude)
mass.m	Satellite mass	4.0	Kilogram	UNCONFIRMED · adcs_ref_c1
mass.imax	Largest principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$
mass.iint	Intermediate principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal
mass.imin	Smallest principal inertia	0.0067	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.10^2)$

6. Design loop per case: every configuration compared

ais_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (feasible)

Converged after 6 iteration(s); knowledge class **coarse**; sensors: magnetometer, sun_sensors, gyro, gnss, earth_sensor. Demand on the POP orbit: peak disturbance $5.634e-07$ N m (sun), stored momentum requirement 0.001466 N m s, torque requirement $5.757e-05$ N m, weakest field 19.76 μ T, detumble momentum 0.00733 N m s.

iter	knobs	options feasible	selection	changes for the next iteration	blocked
1	laws as written	2/25	mtq (closest (not feasible))	nadir_pointing/fmr+mtq: power -> electromagnetic pump with more copper (λ 0.1 -> 0.3 kg/W) nadir_pointing/mtq: thin margin (9.04 of 10): fly every mtq_pointing algorithm sun_acquisition/mtq: fly every sun_acquisition algorithm sun_referencing/mtq: fly every mtq_pointing algorithm	16
2	pump λ 0.3 kg/W	5/25	mtq_fmr (feasible)	nadir_pointing/mtq: thin margin (9.04 of 10): tune every mtq_pointing law's gains (Bruni & Celani 2017) sun_acquisition/mtq: tune every sun_acquisition law's gains, min over the gains of the worst seed (Bruni & Celani 2017) sun_referencing/mtq: tune every mtq_pointing law's gains, min over the gains of the worst seed (Bruni & Celani 2017)	18
3	pump λ 0.3 kg/W	5/25	mtq_fmr (feasible)	mtqp: authority $\times 1 \rightarrow \times 1.5$ (performance)	18
4	mtqp $\times 1.5$, pump λ 0.3 kg/W	6/25	mtq_fmr (feasible)	mtqp: authority $\times 1.5 \rightarrow \times 2.25$ (performance)	18
5	mtqp $\times 2.25$, pump λ 0.3 kg/W	5/25	mtq_fmr (feasible)	mtqp: authority back to $\times 1.5$ (no improvement)	19
6	mtqp $\times 1.5$, pump λ 0.3 kg/W	6/25	mtq_fmr (feasible)	—	19

Why the loop stopped (nothing left that a knob can change):

- detumble/rcs: power fails at the sized authority (rcs); the part's standby power is the floor
- mtqp: more authority did not reduce the performance violation ($16.2 \rightarrow 16.2$); kept at $\times 1.5$
- nadir_pointing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump
- nadir_pointing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor
- nadir_pointing/rw+rcs: power fails at the sized authority (rw); the part's standby power is the floor
- nadir_pointing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- nadir_pointing/vscmg+rcs: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_acquisition/cmg: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_acquisition/rw: power fails at the sized authority (rw); the part's standby power is the floor
- sun_acquisition/vscmg: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_referencing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor

- sun_referencing/fmr+rscs: power is the thrusters' valve power (RCS dumping), not the pump
- sun_referencing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor
- sun_referencing/rw+rscs: power fails at the sized authority (rw); the part's standby power is the floor
- sun_referencing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/vscmg+rscs: power fails at the sized authority (vscmg); the part's standby power is the floor

Every configuration, ours and the benchmarks, flown and scored the same way.

Selection rule (node `select`): least mass_kg, then power_W, then volume_L, then

simplicity among feasible solution families. Selected: `mtq_fmr`; the benchmarks

ranked by the same rule give `mtq_rw` (closest (not feasible)).

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
<code>mtq_fmr</code>	solution	1	pass	1.45	2.54	0.537	fluid loop, 3 rings (ours)	—
<code>mtq</code>	solution	—	FAIL	0.258	2.28	0.0274	coils only	sun_referenc sun_ape_p99 (performance)
<code>mtq_fmr_rcs</code>	solution	—	FAIL	1.96	2.59	1.52	Sized N2O cold-gas RCS, 6 couples ×1 fluid loop, 3 rings (ours)	budget: mass_kg 1.9 1.6; budget: volume_L 1.5 1
<code>mtq_rw</code>	benchmark	—	FAIL	0.393	2.73	0.0703	CubeSpace CubeWheel CW0017 ×3	sun_referenc power_mean (power)
<code>mtq_rw_rcs</code>	benchmark	—	FAIL	0.911	2.78	1.05	CubeSpace CubeWheel CW0017 ×3 Sized N2O cold-gas RCS, 6 couples ×1	sun_referenc power_mean (power); bud volume_L 1.0 1
<code>mtq_cmeg</code>	benchmark	—	FAIL	1.41	5.83	0.809	Tensor Tech ADCS400 ×4	sun_referenc power_mean (power)
<code>mtq_vscmg</code>	benchmark	—	FAIL	1.41	5.83	0.809	Tensor Tech ADCS400 ×4	sun_referenc power_mean (power)
<code>mtq_cmeg_rcs</code>	benchmark	—	FAIL	1.93	5.88	1.79	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_referenc power_mean (power); bud mass_kg 1.9 1.6; budget: volume_L 1.7 1

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
mtq_vscmg_rcs	benchmark	—	FAIL	1.93	5.88	1.79	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_referenc power_mean (power); bud mass_kg 1.9 1.6; budget: volume_L 1.7 1

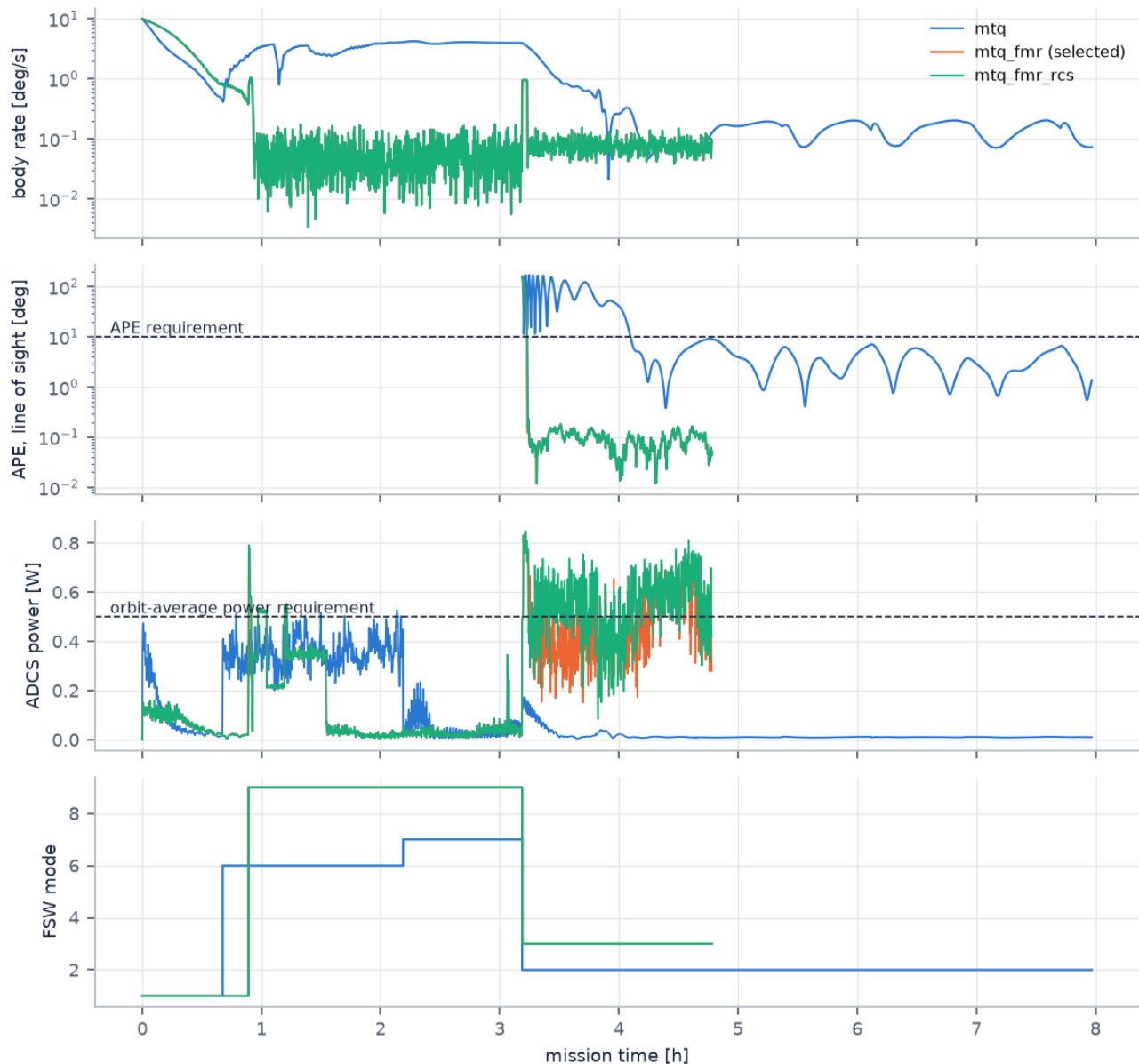
Selected method per mission mode:

mode	option	feasible	objective (worst seed)	algorithms
detumble	mtq	pass	41.01 detumble_time	attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2
sun_acquisition	fmr	pass	2.292 sun_acquisition_time	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
sun_referencing	fmr+mtq	pass	0.3797 sun_ape_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
nadir_pointing	fmr+mtq	pass	0.1471 ape_los_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2

Every one of our solution families flown as the full mission (detumble → Sun acquisition → nadir), with its best methods from the loop. Whether or not it is selected, this is how it behaves (node family_missions):

family	feasible	mass [kg]	methods	detumble [min]	APE [deg]	AKE [deg]	mean power [W]	C = Rust
mtq	FAIL	0.258	detumble=mtq, sun_acquisition=mtq, sun_referencing=mtq, nadir_pointing=mtq	40.46 pass MC 100 % pass	6.626 pass MC 0 % pass	0.1437 pass MC 100 % pass	0.09055 pass MC 100 % pass	pass
mtq_fmr (selected)	pass	1.45	detumble=mtq, sun_acquisition=fmr, sun_referencing=fmr+mtq, nadir_pointing=fmr+mtq	53.46 pass MC 100 % pass	0.1624 pass MC 100 % pass	0.1605 pass MC 100 % pass	0.2223 pass MC 100 % pass	pass
mtq_fmr_rcs	FAIL	1.96	detumble=mtq, sun_acquisition=fmr, sun_referencing=fmr+mtq, nadir_pointing=fmr+mtq	53.46 pass MC 100 % pass	0.1624 pass MC 100 % pass	0.1605 pass MC 100 % pass	0.2514 pass MC 100 % pass	pass

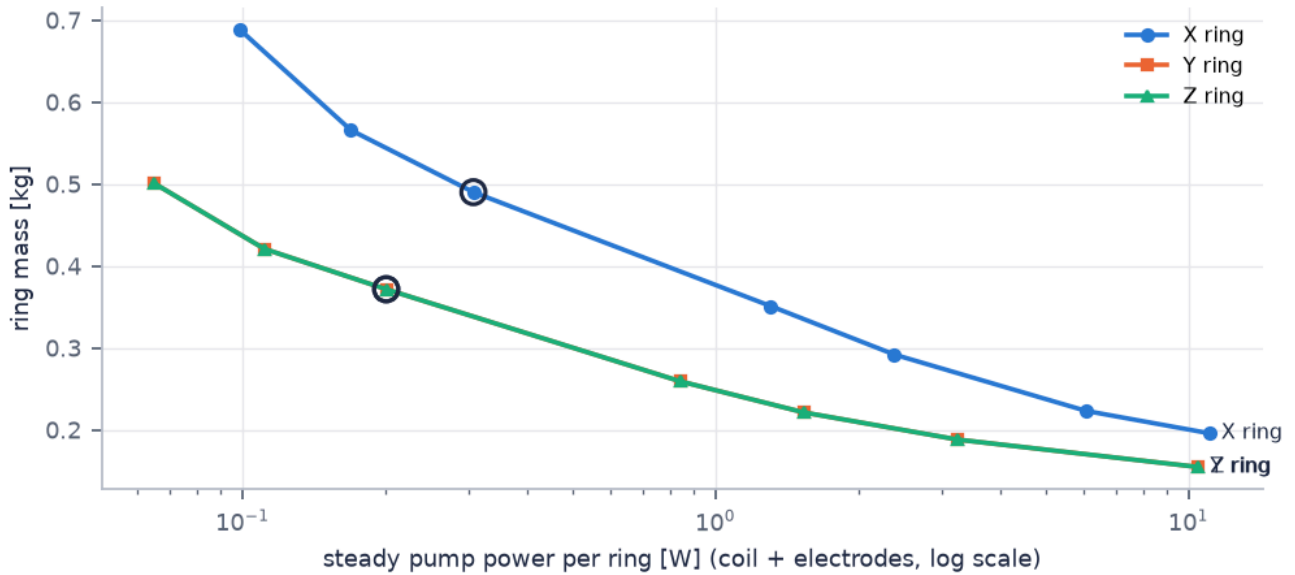
ais_3u: every solution family flown as the dispatched mission (C flight software)



ais_3u: body rate, pointing error, power and flight-software mode through the mission, one line per family.

Coils only (mtq), how it behaves. It is kept as designed; coils-only nadir pointing is to be improved later from the owner's references.

- **Detumble:** 41.01 min in the mode test and 40.46 min in the mission (requirement 284 min); it passes in 100 % of the Monte Carlo runs.
- **Sun acquisition** (Sun-spin, sunspin_damped@spin_rate_dps=4,ss_gain=0.3): the coils spin the body up about the Sun line and bring the power face towards the Sun, so coils alone are largely sufficient here. They do not meet the case's line, though: the power face within 20 deg after 82.32 min (requirement 95 min), and a 95th-percentile Sun angle of 15.5 deg over the last half orbit (limit 20 deg).
- **Sun referencing:** 85.94 deg (p99.73) with mtq_celani2015@mtq_gain_p=0.25,mtq_gain_d=1,handover_out_dps=0.25; the coils cannot hold a three-axis Sun attitude.
- **Nadir pointing** is where the coils fall short. From a settled start the best law (mtq_celani2026@mtq_gain_p=4,mtq_gain_d=0.25,handover_out_dps=0.5) holds the line of sight to 6.278 deg (p99.73) against 10 deg. In the full mission nadir is commanded while the Sun-spin is still turning, and the coils cannot take that momentum out in time: 6.626 deg.



Electromagnetic pump design, ais_3u: each ring's mass against steady power as the mass/power rate λ runs from 0.003 to 3 kg/W; circles mark the converged designs ($\lambda = 0.3$ kg/W).

Monte Carlo of the dispatched mission (12 runs; inertia, residual_dipole, cm_offset, solar_flux, kp, accommodation, reflectivity, arg_lat_deg):

metric	req	mean ± std	range	pass rate
detumble_time	284	56.87 ± 22.8	[7.642, 92.08]	100 %
ape_los_p9973	10	0.257 ± 0.0696	[0.1434, 0.3766]	100 %
ake_los_p9973	5	0.2535 ± 0.0691	[0.14, 0.3726]	100 %
power_mean	0.5	0.3364 ± 0.0943	[0.2279, 0.4991]	100 %
power_peak	—	0.8564 ± 0.0112	[0.8365, 0.8717]	—
propellant	—	0 ± 0	[0, 0]	—

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

metric	req	SILS	soft OILS, C	soft OILS, Rust
detumble_time	284	53.46 pass	53.46 pass	53.46 pass
ape_los_p9973	10	0.1624 pass	0.1467 pass	0.1402 pass
ake_los_p9973	5	0.1605 pass	0.1468 pass	0.1413 pass
power_mean	0.5	0.2223 pass	0.2048 pass	0.2027 pass
power_peak	—	0.858 —	0.8615 —	0.8622 —
propellant	—	0 —	0 —	0 —

ais_img_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (feasible)

Converged after 6 iteration(s); knowledge class **fine**; sensors: star_tracker, magnetometer, sun_sensors, gyro, gnss, earth_sensor. Demand on the POP orbit: peak disturbance 4.367e-07 N m (sun), stored momentum requirement 0.001553 N m s, torque requirement 5.757e-05 N m, weakest field 23.62 μ T, detumble momentum 0.00733 N m s.

iter	knobs	options feasible	selection	changes for the next iteration	blocked
1	laws as written	9/25	mtq_fmr (closest (not feasible))	nadir_pointing/fmr+mtq: fly every pointing algorithm nadir_pointing/fmr+rcs: fly every pointing algorithm nadir_pointing/mtq: fly every mtq_pointing algorithm sun_acquisition/mtq: fly every sun_acquisition algorithm sun_referencing/mtq: fly every mtq_pointing algorithm	15
2	laws as written	9/25	mtq_fmr (closest (not feasible))	rate stability: gyro noise x1 -> x0.3 (fibre-optic class) nadir_pointing/mtq: tune every mtq_pointing law's gains, min over the gains of the worst seed (Bruni & Celani 2017) sun_acquisition/mtq: tune every sun_acquisition law's gains, min over the gains of the worst seed (Bruni & Celani 2017) sun_referencing/mtq: tune every mtq_pointing law's gains, min over the gains of the worst seed (Bruni & Celani 2017)	14
3	gyro noise x0.3	9/25	mtq_fmr (closest (not feasible))	gyro grade back to x1: rate-stability violation 13.6 -> 13.7 nadir_pointing/fmr+mtq: fluid-loop flow sensor 2 -> 0.5 mm/s (1 sigma) mtqp: authority x1 -> x1.5 (performance)	13
4	mtqp x1.5, flow sensor 0.5 mm/s	10/25	mtq_fmr (closest (not feasible))	mtqp: authority back to x1 (no improvement)	13
5	mtqp x1, flow sensor 0.5 mm/s	10/25	mtq_fmr (closest (not feasible))	mass (mtq_fmr): one star-tracker head instead of two	13
6	mtqp x1, 1 ST head, flow sensor 0.5 mm/s	10/25	mtq_fmr (feasible)	—	13

Why the loop stopped (nothing left that a knob can change):

- mtqp: more authority did not reduce the performance violation ($2.6e+03 \rightarrow 2.85e+03$); kept at x1
- nadir_pointing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump
- nadir_pointing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- nadir_pointing/vscmg+rcs: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_acquisition/cmg: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_acquisition/vscmg: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_referencing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_referencing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump
- sun_referencing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/vscmg+rcs: power fails at the sized authority (vscmg); the part's standby power is the floor

Every configuration, ours and the benchmarks, flown and scored the same way.

Selection rule (node `select`): least mass_kg, then power_W, then volume_L, then simplicity among feasible solution families. Selected: `mtq_fmr`; the benchmarks ranked by the same rule give `mtq_rw` (feasible).

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
mtq_fmr	solution	1	pass	1.43	5.88	0.534	fluid loop, 3 rings (ours)	—
mtq	solution	—	FAIL	0.53	2.9	0.0118	coils only	sun_acquisiti sun_acquisiti (performance sun_angle_p (performance sun_referenc sun_ape_p99 (performance nadir_pointin ape_los_p99 (performance rate_stability (performance
mtq_fmr_rcs	solution	—	FAIL	1.89	5.93	1.37	Sized N2O cold-gas RCS, 6 couples ×1 fluid loop, 3 rings (ours)	budget: mass 1.89 > 1.6; b volume_L 1.3
mtq_rw	benchmark	1	pass	0.696	3.66	0.0676	CubeSpace CubeWheel CW0017 ×3	—
mtq_rw_rcs	benchmark	2	pass	1.16	3.71	0.908	CubeSpace CubeWheel CW0017 ×3 Sized N2O cold-gas RCS, 6 couples ×1	—
mtq_cmg	benchmark	—	FAIL	1.72	6.76	0.806	Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 1.7
mtq_vscmg	benchmark	—	FAIL	1.72	6.76	0.806	Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 1.7

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
mtq_cmg_rcs	benchmark	—	FAIL	2.18	6.81	1.65	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 2.1 budget: volur 1.65 > 1
mtq_vscmg_rcs	benchmark	—	FAIL	2.18	6.81	1.65	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 2.1 budget: volur 1.65 > 1

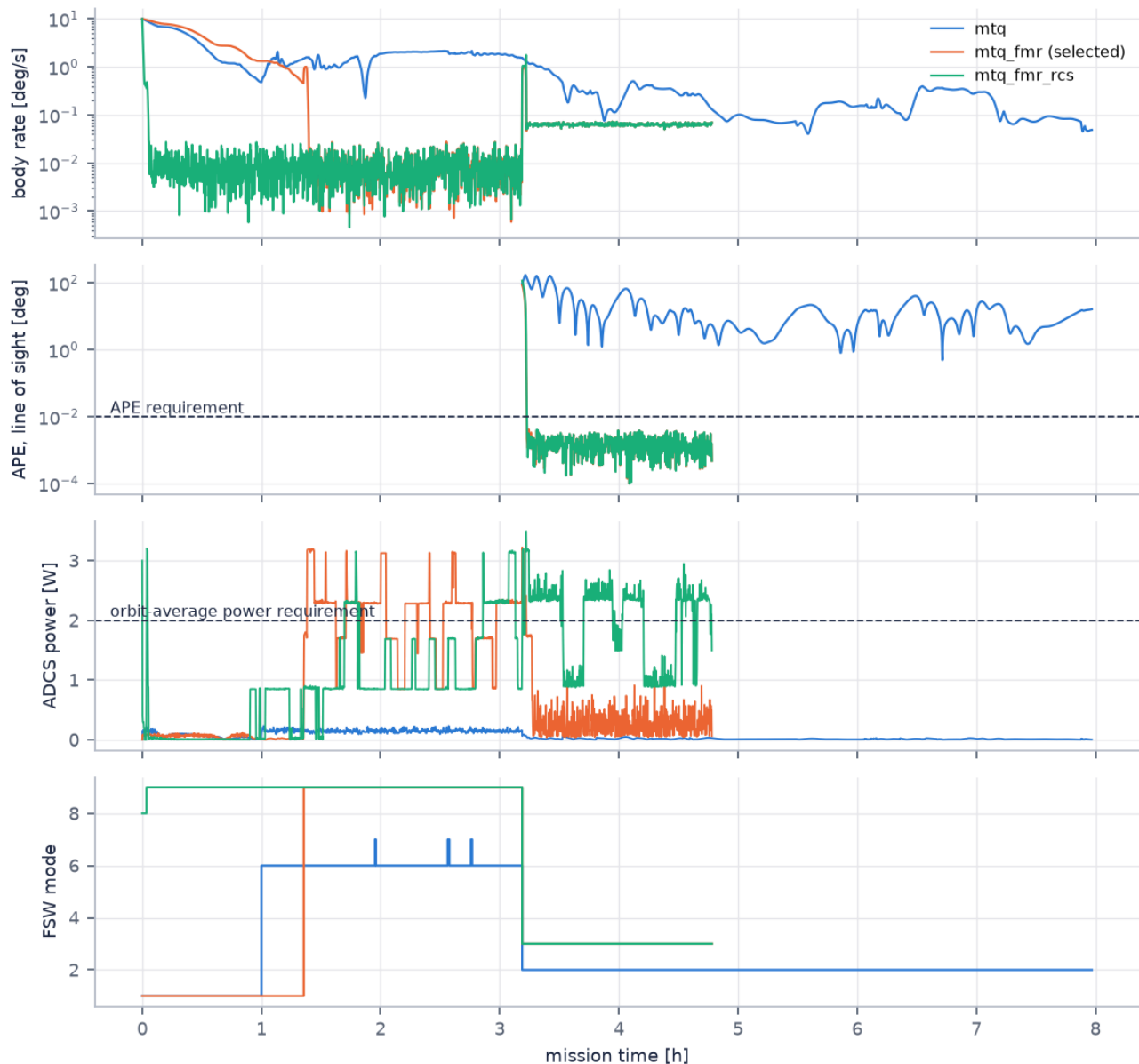
Selected method per mission mode:

mode	option	feasible	objective (worst seed)	algorithms
detumble	mtq	pass	58.91 detumble_time	attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2
sun_acquisition	fmr	pass	2.292 sun_acquisition_time	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
sun_referencing	fmr+mtq	pass	0.1566 sun_ape_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
nadir_pointing	fmr+mtq	pass	0.004483 ape_los_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2

Every one of our solution families flown as the full mission (detumble → Sun acquisition → nadir), with its best methods from the loop. Whether or not it is selected, this is how it behaves (node `family_missions`):

family	feasible	mass [kg]	methods	detumble [min]	APE [deg]	AKE [deg]	mean power [W]	C = Rust
mtq	FAIL	0.53	detumble=mtq, sun_acquisition=mtq, sun_referencing=mtq, nadir_pointing=mtq	59.99 pass MC 100 % pass	23.34 FAIL MC 0 % pass	0.01675 FAIL MC 42 % pass	0.0592 pass MC 100 % pass	pass
mtq_fmr (selected)	pass	1.43	detumble=mtq, sun_acquisition=fmr, sun_referencing=fmr+mtq, nadir_pointing=fmr+mtq	81.21 pass MC 100 % pass	0.004249 pass MC 100 % pass	0.003157 pass MC 100 % pass	0.9336 pass MC 100 % pass	pass
mtq_fmr_rcs	FAIL	1.89	detumble=rscs, sun_acquisition=fmr, sun_referencing=fmr+mtq, nadir_pointing=fmr+mtq	1.075 pass MC 100 % pass	0.004241 pass MC 100 % pass	0.003154 pass MC 100 % pass	1.219 pass MC 100 % pass	pass

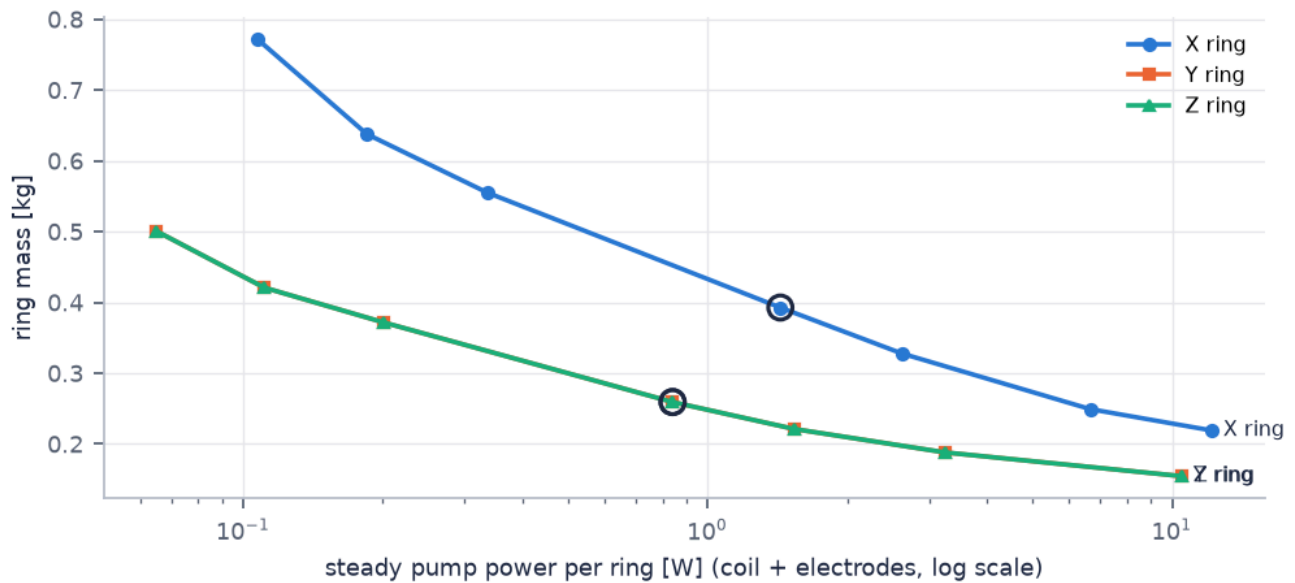
ais_img_3u: every solution family flown as the dispatched mission (C flight software)



ais_img_3u: body rate, pointing error, power and flight-software mode through the mission, one line per family.

Coils only (mtq), how it behaves. It is kept as designed; coils-only nadir pointing is to be improved later from the owner's references.

- **Detumble:** 58.91 min in the mode test and 59.99 min in the mission (requirement 284 min); it passes in 100 % of the Monte Carlo runs.
- **Sun acquisition** (Sun-spin, sunspin_deruiter2011@spin_rate_dps=2,ss_gain=3): the coils spin the body up about the Sun line and bring the power face towards the Sun, so coils alone are largely sufficient here. They do not meet the case's line, though: the power face within 20 deg after 34.49 min (requirement 95 min), and a 95th-percentile Sun angle of 90.77 deg over the last half orbit (limit 20 deg).
- **Sun referencing:** 19.26 deg (p99.73) with mtq_celani2026@mtq_gain_p=4,mtq_gain_d=0.25,handover_out_dps=0.25; the coils cannot hold a three-axis Sun attitude.
- **Nadir pointing** is where the coils fall short. From a settled start the best law (mtq_celani2026@mtq_gain_p=4,mtq_gain_d=0.25,handover_out_dps=0.5) holds the line of sight to 26.78 deg (p99.73) against 0.01 deg. In the full mission nadir is commanded while the Sun-spin is still turning, and the coils cannot take that momentum out in time: 23.34 deg.



Electromagnetic pump design, ais_img_3u: each ring's mass against steady power as the mass/power rate λ runs from 0.003 to 3 kg/W; circles mark the converged designs ($\lambda = 0.1$ kg/W).

Monte Carlo of the dispatched mission (12 runs; inertia, residual_dipole, cm_offset, solar_flux, accommodation, reflectivity, arg_lat_deg):

metric	req	mean $\pm$ std	range	pass rate
detumble_time	284	72.62 $\pm$ 14.6	[42.96, 90.17]	100 %
ape_los_p9973	0.01	0.004304 $\pm$ 0.000387	[0.003872, 0.005259]	100 %
ake_los_p9973	0.005	0.003296 $\pm$ 0.000449	[0.002766, 0.0045]	100 %
power_mean	2	0.8959 $\pm$ 0.351	[0.4424, 1.624]	100 %
power_peak	—	3.224 $\pm$ 0.00874	[3.214, 3.238]	—
propellant	—	0 $\pm$ 0	[0, 0]	—

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

metric	req	SILS	soft OILS, C	soft OILS, Rust
detumble_time	284	81.21 pass	81.12 pass	81.12 pass
ape_los_p9973	0.01	0.004249 pass	0.00452 pass	0.004579 pass
ake_los_p9973	0.005	0.003157 pass	0.003185 pass	0.003208 pass
power_mean	2	0.9336 pass	0.8047 pass	0.7481 pass
power_peak	—	3.22 —	3.223 —	3.222 —
propellant	—	0 —	0 —	0 —

7. Magnetorquer-only literature

Sixteen magnetorquer-only papers were reviewed (literature_review_v2.pptx). Every law that the accessible text specifies is now in the flight software, in C and Rust (bit for bit identical), in the pseudocode, and in the registry as a candidate of its slot: Lovera & Astolfi 2004, Celani 2015, Avanzini et al. 2021, Celani 2026 (boresight, for the payload and for the power face), TANGO's frozen-Riccati LQR (flown), de Ruiter 2011, and the P11 → P5 chain already in place (UPMSat-2 spin-up, He et al. Sun spin). Bruni & Celani's min–max tuning is node `tune`; Celani's Floquet certificate is node `certify`. The paper-by-paper record, including what was not implemented and why, is `docs/MTQ_LITERATURE.md`.

The Sun-spin → nadir hand-over

The coils-only mission commands nadir from a body spinning at about 4 °/s. The magnetic states now despin first. Above 1 °/s of rate error they run the B-dot law on that error with the detumble's optimal gain (Avanzini & Giulietti 2012). The spin-up's high gain only drags the rate along the turning field. The pointing law takes over below 0.5 °/s, held for 60 s. The magnetic capture then takes about 1.5 orbits, so node `dispatch` gives a coils-only nadir three orbits after the command, and the metrics take the last half orbit. The coils-only nadir mode test starts the same way (arbitrary attitude, 6 °/s), and node `tune` flies the hand-over exit threshold with the gains, so the dispatched gains capture as well as hold. Coils-only mission nadir APE (p99.73): `ais_3u` 6.626° **pass**; `ais_img_3u` 23.34° **FAIL**. `ais_3u` Monte Carlo: 0 % of runs within 10° (13.76–46.16°). `ais_img_3u` Monte Carlo: 0 % of runs within 0.01° (7.426–73.05°).

Why Sun referencing is worse than nadir with coils only

The coils are not short of dipole. The difference is the gravity gradient. At nadir on `ais_3u` the long, minimum-inertia axis is the payload axis, at the gradient's equilibrium, so the gradient is a restoring stiffness (`ais_img_3u` flies its long axis along track, pitch is unstable, and the nadir state cancels the gradient too). Under Sun referencing the attitude is inertial, so the gradient becomes a forcing at twice the orbit rate. A magnetic torque is perpendicular to B, so the part of that forcing along B cannot be rejected at that instant. The Sun state now cancels the modelled gradient (`mtq_gg_ff`); on `ais_3u` with Celani 2026 the Sun-pointing error went from 80.6° to 66.8°. It is still not the 5° line, which agrees with the flown and published figures (TANGO 16°, Celani 2026 24°). The coils-only power attitude remains the Sun spin.

`ais_3u`: coils only, every law at its best gains

mode	law	paper	best gains	feasible	objective (worst seed)
sun_acquisition	<code>sunspin_damped</code>	P11 -> P5, R_z floor	spin_rate_dps 4, ss_gain 0.3	pass	82.32 sun_acquisition_time
sun_acquisition	<code>sunspin_l1l2_e2</code>	P11 -> P5, eclipse E2	spin_rate_dps 4, ss_gain 0.3	FAIL	95.77 sun_acquisition_time

mode	law	paper	best gains	feasible	objective (worst seed)
sun_acquisition	sunspin_deruiter2011	P11 -> P2 de Ruiter 2011	nominal	FAIL	30.19 sun_acquisition_time
sun_acquisition	sunspin_l1l2	P11 UPMSat-2 -> P5 He et al. 2023	spin_rate_dps 2, ss_gain 0.3	FAIL	67.11 sun_acquisition_time
sun_acquisition	sun_boresight_celani2026	P8 Celani 2026	nominal	FAIL	79.72 sun_acquisition_time
sun_referencing	mtq_celani2015	P4 Celani 2015	mtq_gain_p 0.25, mtq_gain_d 1, handover_out_dps 0.25	FAIL	85.94 sun_ape_p9973
sun_referencing	mtq_avanzini2021	P16 Avanzini et al. 2021	mtq_gain_p 0.25, mtq_gain_d 1, handover_out_dps 0.25	FAIL	85.94 sun_ape_p9973
sun_referencing	mtq_celani2026	P8 Celani 2026	mtq_gain_p 0.25, mtq_gain_d 1, handover_out_dps 0.25	FAIL	86.38 sun_ape_p9973
sun_referencing	mtq_tango2013	P3 TANGO 2013 (flown)	mtq_gain_p 0.25, mtq_gain_d 1, handover_out_dps 0.25	FAIL	135.4 sun_ape_p9973
sun_referencing	mtq_lovera2004	P1 Lovera & Astolfi 2004	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	136.1 sun_ape_p9973
sun_referencing	mtq_pd	baseline PD	nominal	FAIL	151.2 sun_ape_p9973
sun_referencing	mtq_rate_damp	baseline rate damping	nominal	FAIL	162.8 sun_ape_p9973
sun_referencing	mtq_lqr	baseline LQR	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	173.6 sun_ape_p9973
sun_referencing	mtq_smc	baseline SMC	nominal	FAIL	179 sun_ape_p9973
nadir_pointing	mtq_celani2026	P8 Celani 2026	mtq_gain_p 4, mtq_gain_d 0.25, handover_out_dps 0.5	pass	6.278 ape_los_p9973
nadir_pointing	mtq_pd	baseline PD	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 1	pass	7.989 ape_los_p9973

mode	law	paper	best gains	feasible	objective (worst seed)
nadir_pointing	mtq_smc	baseline SMC	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	12 ape_los_p9973
nadir_pointing	mtq_tango2013	P3 TANGO 2013 (flown)	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 1	FAIL	23.86 ape_los_p9973
nadir_pointing	mtq_celani2015	P4 Celani 2015	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 1	FAIL	27.16 ape_los_p9973
nadir_pointing	mtq_lovera2004	P1 Lovera & Astolfi 2004	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	34.16 ape_los_p9973
nadir_pointing	mtq_lqr	baseline LQR	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	49.52 ape_los_p9973
nadir_pointing	mtq_avanzini2021	P16 Avanzini et al. 2021	mtq_gain_p 4, mtq_gain_d 1, handover_out_dps 0.5	FAIL	68.8 ape_los_p9973
nadir_pointing	mtq_rate_damp	baseline rate damping	nominal	FAIL	175 ape_los_p9973

ais_3u: Floquet multipliers of the coils-only nadir loop (linearised with gyroscopic and gravity-gradient terms, coil duty and the field along one orbit); all $|\mu| < 1$ certifies the periodic loop.

law	gains	max $ \mu $	certified
mtq_pd	nominal	0.5435	pass
mtq_rate_damp	nominal	1.0000	pass
mtq_lovera2004	nominal	1.1470	FAIL
mtq_celani2015	nominal	0.7831	pass
mtq_avanzini2021	nominal	2.5192	FAIL
mtq_celani2026 (dispatched)	{'mtq_gain_p': 4.0, 'mtq_gain_d': 0.25}	0.9128	pass
mtq_tango2013	nominal	0.0702	pass

ais_img_3u: coils only, every law at its best gains

mode	law	paper	best gains	feasible	objective (worst seed)
sun_acquisition	sunspin_deruiter2011	P11 -> P2 de Ruiter 2011	spin_rate_dps 2, ss_gain 3	FAIL	34.49 sun_acquisition_time

mode	law	paper	best gains	feasible	objective (worst seed)
sun_acquisition	sunspin_l1l2	P11 UPMSat- 2 -> P5 He et al. 2023	spin_rate_dps 4, ss_gain 3	FAIL	66.71 sun_acquisition_time
sun_acquisition	sunspin_l1l2_e2	P11 -> P5, eclipse E2	spin_rate_dps 4, ss_gain 3	FAIL	66.71 sun_acquisition_time
sun_acquisition	sunspin_damped	P11 -> P5, R_z floor	spin_rate_dps 2, ss_gain 1	FAIL	78.51 sun_acquisition_time
sun_acquisition	sun_boresight_celani2026	P8 Celani 2026	nominal	FAIL	135.2 sun_acquisition_time
sun_referencing	mtq_celani2026	P8 Celani 2026	mtq_gain_p 4, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	19.26 sun_ape_p9973
sun_referencing	mtq_tango2013	P3 TANGO 2013 (flown)	mtq_gain_p 0.25, mtq_gain_d 4, handover_out_dps 0.25	FAIL	58.1 sun_ape_p9973
sun_referencing	mtq_celani2015	P4 Celani 2015	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	66.54 sun_ape_p9973
sun_referencing	mtq_avanzini2021	P16 Avanzini et al. 2021	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	66.54 sun_ape_p9973
sun_referencing	mtq_lovera2004	P1 Lovera & Astolfi 2004	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	136.8 sun_ape_p9973
sun_referencing	mtq_rate_damp	baseline rate damping	nominal	FAIL	153 sun_ape_p9973
sun_referencing	mtq_pd	baseline PD	nominal	FAIL	171 sun_ape_p9973
sun_referencing	mtq_smc	baseline SMC	nominal	FAIL	173.2 sun_ape_p9973
sun_referencing	mtq_lqr	baseline LQR	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	178 sun_ape_p9973
nadir_pointing	mtq_celani2026	P8 Celani 2026	mtq_gain_p 4, mtq_gain_d 0.25, handover_out_dps 0.5	FAIL	26.78 ape_los_p9973

mode	law	paper	best gains	feasible	objective (worst seed)
nadir_pointing	mtq_celani2015	P4 Celani 2015	mtq_gain_p 1, mtq_gain_d 1, handover_out_dps 0.25	FAIL	52.39 ape_los_p9973
nadir_pointing	mtq_lovera2004	P1 Lovera & Astolfi 2004	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.5	FAIL	76.26 ape_los_p9973
nadir_pointing	mtq_tango2013	P3 TANGO 2013 (flown)	mtq_gain_p 1, mtq_gain_d 1, handover_out_dps 0.25	FAIL	77.66 ape_los_p9973
nadir_pointing	mtq_rate_damp	baseline rate damping	nominal	FAIL	123.5 ape_los_p9973
nadir_pointing	mtq_avanzini2021	P16 Avanzini et al. 2021	nominal	FAIL	134.3 ape_los_p9973
nadir_pointing	mtq_pd	baseline PD	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 0.25	FAIL	150.8 ape_los_p9973
nadir_pointing	mtq_lqr	baseline LQR	nominal	FAIL	166.8 ape_los_p9973
nadir_pointing	mtq_smc	baseline SMC	mtq_gain_p 0.25, mtq_gain_d 0.25, handover_out_dps 1	FAIL	176.3 ape_los_p9973

ais_img_3u: Floquet multipliers of the coils-only nadir loop (linearised with gyroscopic and gravity-gradient terms, coil duty and the field along one orbit); all $|\mu| < 1$ certifies the periodic loop.

law	gains	max $ \mu $	certified
mtq_pd	nominal	4.9033	FAIL
mtq_rate_damp	nominal	1.9820	FAIL
mtq_lovera2004	nominal	4.8218	FAIL
mtq_celani2015	nominal	2.4125	FAIL
mtq_avanzini2021	nominal	7.0389	FAIL
mtq_celani2026 (dispatched)	{'mtq_gain_p': 4.0, 'mtq_gain_d': 0.25}	2.7519	FAIL
mtq_tango2013	nominal	7.1476	FAIL

8. SILS scenario verification

Every scenario on the Rust engine with the C flight software behind the byte HAL (POP in the loop). 25 of 40 scenarios meet every requirement they carry; the failures are findings of the design (section 10), not engine errors: the same verdicts come out of the MATLAB twin (section 7).

scenario	what	product	duration [s]	requirement metrics: value / req
agile_slew_cmgy	3U imaging — (cmgy): 90 deg pitch in 15 s	TRN-P-3U-CMG	600	ape_los_on_target_p9973 0.004544 / 0.01 pass
agile_slew_fmr_rcs	3U imaging — (fmr_rcs): 90 deg pitch in 15 s	TRN-P-3U-FMR-RCS	600	ape_los_on_target_p9973 0.006614 / 0.01 pass
agile_slew_img	3U imaging — (img): 90 deg pitch in 15 s	TRN-P-3U-IMG	600	ape_los_on_target_p9973 0.008686 / 0.01 pass
agile_slew_rw_rcs	3U imaging — (rw_rcs): 90 deg pitch in 15 s	TRN-P-3U-RW-RCS	600	ape_los_on_target_p9973 0.008702 / 0.01 pass
agile_slew_vscmg	3U imaging — (vscmg): 90 deg pitch in 15 s	TRN-P-3U-VSCMG	600	ape_los_on_target_p9973 0.01008 / 0.01 FAIL
detumble_ais	3U AIS — B-dot detumble from a 10 deg/s tumble (coils only)	TRN-P-3U-AIS	1.722e+04	detumble_time 37.79 / 284 pass power_mean 0.0217 / 0.5 pass power_peak 0.5502 / 1.5 pass
detumble_ais_bangbang	3U AIS — detumble, B-dot law bangbang	TRN-P-3U-AIS	1.722e+04	detumble_time 54.02 / 284 pass power_mean 0.4366 / 0.5 pass power_peak 0.717 / 1.5 pass
detumble_ais_mag	3U AIS — detumble, B-dot law mag	TRN-P-3U-AIS	1.722e+04	detumble_time 41.16 / 284 pass power_mean 0.07839 / 0.5 pass power_peak 0.6377 / 1.5 pass
detumble_img	3U imaging — B-dot detumble from a 10 deg/s tumble	TRN-P-3U-IMG	1.722e+04	detumble_time 47.54 / 284 pass
fault_coil_ais	3U AIS — X coil fails during detumble	TRN-P-3U-AIS	1.722e+04	detumble_time 82.17 / 284 pass power_mean 0.01821 / 0.5 pass power_peak 0.5502 / 1.5 pass
fault_gimbal_cmgy	3U imaging — one CMG gimbal sticks during fine hold	TRN-P-3U-CMG	5740	ape_los_p9973 0.005509 / 0.01 pass ake_los_p9973 0.002624 / 0.005 pass rate_stability_p9973 0.002963 / 0.005 pass power_mean 1.61 / 2 pass
fault_gyro_ais	3U AIS — gyro bias step of 0.1 deg/s during nadir hold	TRN-P-3U-AIS	1.722e+04	ape_los_last_orbit 52.78 / 10 FAIL ake_los_last_orbit 41.92 / 5 FAIL power_mean 0.01056 / 0.5 pass

scenario	what	product	duration [s]	requirement metrics: value / req
fault_st_img	3U imaging — star-tracker head 2 fails during fine hold	TRN-P-3U-IMG	5740	ape_los_p9973 0.1022 / 0.01 FAIL ake_los_p9973 0.1093 / 0.005 FAIL rate_stability_p9973 0.003771 / 0.005 pass power_mean 1.557 / 2 pass
fault_wheel_img	3U imaging — reaction wheel 1 fails during fine hold (FDIR + coils)	TRN-P-3U-IMG	5740	ape_los_p9973 28.62 / 0.01 FAIL ake_los_p9973 0.002858 / 0.005 pass rate_stability_p9973 0.6244 / 0.005 FAIL power_mean 1.213 / 2 pass
fine_hold_cmrg	3U imaging — fine nadir hold, coils + 4 SGCMG pyramid	TRN-P-3U-CMG	5740	ape_los_p9973 0.005021 / 0.01 pass ake_los_p9973 0.002626 / 0.005 pass rate_stability_p9973 0.002437 / 0.005 pass power_mean 1.61 / 2 pass
fine_hold_fmr	3U imaging — fine nadir hold, coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	5740	ape_los_p9973 0.005689 / 0.01 pass ake_los_p9973 0.002631 / 0.005 pass rate_stability_p9973 0.004729 / 0.005 pass power_mean 0.08406 / 2 pass
fine_hold_fmr_rcs	3U imaging — fine nadir hold, coils + fluid rings + cold-gas RCS	TRN-P-3U-FMR-RCS	5740	ape_los_p9973 0.005689 / 0.01 pass ake_los_p9973 0.002631 / 0.005 pass rate_stability_p9973 0.004729 / 0.005 pass power_mean 0.08406 / 2 pass
fine_hold_img	3U imaging — fine nadir hold, reaction wheels + star tracker (0.01 deg)	TRN-P-3U-IMG	5740	ape_los_p9973 0.01049 / 0.01 FAIL ake_los_p9973 0.002608 / 0.005 pass rate_stability_p9973 0.003552 / 0.005 pass power_mean 1.557 / 2 pass
fine_hold_img_lqr	3U imaging — fine nadir hold, LQR controller	TRN-P-3U-IMG	5740	ape_los_p9973 0.004156 / 0.01 pass ake_los_p9973 0.002652 / 0.005 pass rate_stability_p9973 0.006704 / 0.005 FAIL power_mean 1.558 / 2 pass
fine_hold_img_smc	3U imaging — fine nadir hold, SMC controller	TRN-P-3U-IMG	5740	ape_los_p9973 0.01945 / 0.01 FAIL ake_los_p9973 0.002649 / 0.005 pass rate_stability_p9973 0.003482 / 0.005 pass power_mean 1.557 / 2 pass

scenario	what	product	duration [s]	requirement metrics: value / req
<code>fine_hold_rw_rcs</code>	3U imaging — fine nadir hold, coils + reaction wheels + cold-gas RCS	TRN-P-3U-RW-RCS	5740	ape_los_p9973 0.03388 / 0.01 FAIL ake_los_p9973 0.002602 / 0.005 pass rate_stability_p9973 0.005641 / 0.005 FAIL power_mean 1.53 / 2 pass
<code>fine_hold_vscmg</code>	3U imaging — fine nadir hold, coils + 4 VSCMG pyramid	TRN-P-3U-VSCMG	5740	ape_los_p9973 0.00828 / 0.01 pass ake_los_p9973 0.002636 / 0.005 pass rate_stability_p9973 0.004762 / 0.005 pass power_mean 2.014 / 2 FAIL
<code>mission_ais</code>	3U AIS — integrated chain: detumble then nadir hold (auto mode switch)	TRN-P-3U-AIS	2.87e+04	detumble_time 37.79 / 284 pass ape_los_last_orbit 8.935 / 10 pass ake_los_last_orbit 1.188 / 5 pass
<code>mission_cmg</code>	3U imaging — tumble, detumble, fine hold: coils + 4 SGCMG pyramid	TRN-P-3U-CMG	1.722e+04	detumble_time 57.01 / 284 pass ape_los_last_half_orbit_p9973 0.005241 / 0.01 pass
<code>mission_fmr</code>	3U imaging — tumble, detumble, fine hold: coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	1.722e+04	detumble_time 61.33 / 284 pass ape_los_last_half_orbit_p9973 0.005954 / 0.01 pass
<code>mission_fmr_rcs</code>	3U imaging — tumble, detumble, fine hold: coils + fluid rings + cold-gas RCS	TRN-P-3U-FMR-RCS	1.722e+04	detumble_time 61.33 / 284 pass ape_los_last_half_orbit_p9973 0.005954 / 0.01 pass
<code>mission_img</code>	3U imaging — integrated chain: detumble then fine nadir hold	TRN-P-3U-IMG	1.722e+04	detumble_time 51.01 / 284 pass ape_los_last_half_orbit_p9973 0.01066 / 0.01 FAIL
<code>mission_rw_rcs</code>	3U imaging — tumble, detumble, fine hold: coils + reaction wheels + cold-gas RCS	TRN-P-3U-RW-RCS	1.722e+04	detumble_time 51.01 / 284 pass ape_los_last_half_orbit_p9973 0.007564 / 0.01 pass
<code>mission_vscmg</code>	3U imaging — tumble, detumble, fine hold: coils + 4 VSCMG pyramid	TRN-P-3U-VSCMG	1.722e+04	detumble_time 57.02 / 284 pass ape_los_last_half_orbit_p9973 0.009595 / 0.01 pass
<code>nadir_hold_ais</code>	3U AIS — nadir hold with magnetorquers only (10 deg requirement)	TRN-P-3U-AIS	1.722e+04	ape_los_last_orbit 12.64 / 10 FAIL ake_los_last_orbit 5.491 / 5 FAIL power_mean 0.01054 / 0.5 pass
<code>nadir_hold_ais_css</code>	3U AIS — nadir hold with coarse cosine sun sensors (albedo)	TRN-P-3U-AIS-CSS	1.722e+04	ape_los_last_orbit 39.44 / 10 FAIL ake_los_last_orbit 39.22 / 5 FAIL power_mean 0.01064 / 0.5 pass
<code>slew_cmg</code>	3U imaging — 30 deg target slew in 60 s, coils + 4 SGCMG pyramid	TRN-P-3U-CMG	600	settle_time_after_slew 12.4 / 20 pass ape_los_on_target_p9973 0.004233 / 0.01 pass
<code>slew_fmr</code>	3U imaging — 30 deg target slew in 60 s, coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	600	settle_time_after_slew 12.9 / 20 pass ape_los_on_target_p9973 0.006529 / 0.01 pass

scenario	what	product	duration [s]	requirement metrics: value / req
slew_fmr_rcs	3U imaging — 30 deg target slew in 60 s, coils + fluid rings + cold-gas RCS	TRN-P- 3U-FMR- RCS	600	settle_time_after_slew 12.9 / 20 pass ape_los_on_target_p9973 0.006529 / 0.01 pass
slew_img	3U imaging — 30 deg target slew in 60 s and settle	TRN-P- 3U-IMG	600	settle_time_after_slew 12.1 / 20 pass ape_los_on_target_p9973 0.007925 / 0.01 pass
slew_img_lqr	3U imaging — 30 deg slew, LQR controller	TRN-P- 3U-IMG	600	settle_time_after_slew 0 / 20 pass ape_los_on_target_p9973 0.004439 / 0.01 pass
slew_img_smc	3U imaging — 30 deg slew, SMC controller	TRN-P- 3U-IMG	600	settle_time_after_slew — / 20 FAIL ape_los_on_target_p9973 0.0178 / 0.01 FAIL
slew_rw_rcs	3U imaging — 30 deg target slew in 60 s, coils + reaction wheels + cold-gas RCS	TRN-P- 3U-RW- RCS	600	settle_time_after_slew 12.1 / 20 pass ape_los_on_target_p9973 0.007922 / 0.01 pass
slew_vscmg	3U imaging — 30 deg target slew in 60 s, coils + 4 VSCMG pyramid	TRN-P- 3U- VSCMG	600	settle_time_after_slew 13.2 / 20 pass ape_los_on_target_p9973 0.01083 / 0.01 FAIL
sun_spin_ais	3U AIS — coils-only Sun acquisition: detumble, spin- up, Sun-pointing spin	TRN-P- 3U-AIS	1.722e+04	sun_spin_entry 142.4 / 95 FAIL sun_angle_last_orbit 93.84 / 20 FAIL spin_rate_error_last_orbit 1.116 / 0.5 FAIL sun_spin_share_last_orbit 13.75 / 90 FAIL power_mean 0.45 / 0.5 pass

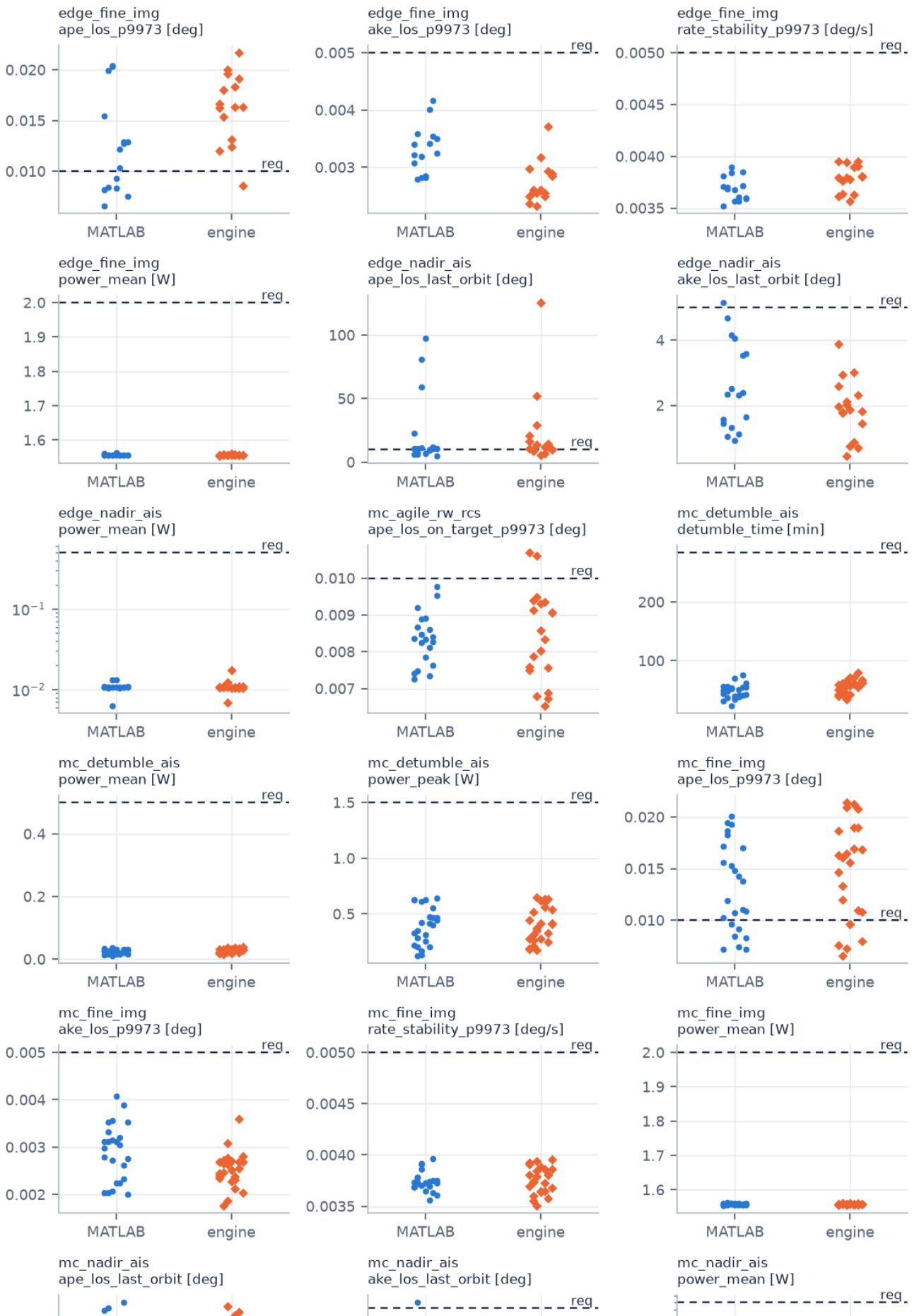
9. Monte Carlo and edge campaigns vs the MATLAB twin

Every campaign of `campaigns/*.toml` flown on the Rust engine with the draws of `asils.campaign.draw` (same dispersions, bounds, run count, per-run seeds), next to the MATLAB twin's campaign. Monte Carlo realisations differ (different random streams) so distributions are compared; edge runs are compared run by run. The engine's flight software keeps the nominal inertia while the plant is dispersed; the MATLAB twin's laws read the dispersed inertia.

campaign	type	metric	req	MATLAB mean $\pm$ std	MATLAB pass	engine mean $\pm$ std	engine pass
edge_fine_img	edge	ape_los_p9973	0.01	0.01236 $\pm$ 0.00475	40 %	0.01625 $\pm$ 0.00351	7 %
edge_fine_img	edge	ake_los_p9973	0.005	0.003307 $\pm$ 0.000418	100 %	0.002739 $\pm$ 0.000362	100 %
edge_fine_img	edge	rate_stability_p9973	0.005	0.003689 $\pm$ 0.000116	100 %	0.003788 $\pm$ 0.00013	100 %
edge_fine_img	edge	power_mean	2	1.558 $\pm$ 0.00216	100 %	1.557 $\pm$ 0.00112	100 %
edge_nadir_ais	edge	ape_los_last_orbit	10	22.16 $\pm$ 28.2	41 %	21.49 $\pm$ 29	29 %
edge_nadir_ais	edge	ake_los_last_orbit	5	2.577 $\pm$ 1.37	94 %	1.899 $\pm$ 0.905	100 %
edge_nadir_ais	edge	power_mean	0.5	0.01074 $\pm$ 0.00143	100 %	0.01096 $\pm$ 0.00194	100 %
mc_agile_rw_rcs	montecarlo	ape_los_on_target_p9973	0.01	0.008337 $\pm$ 0.000712	100 %	0.008427 $\pm$ 0.00124	90 %
mc_detumble_ais	montecarlo	detumble_time	284	48.5 $\pm$ 12	100 %	56.29 $\pm$ 11.4	100 %
mc_detumble_ais	montecarlo	power_mean	0.5	0.02267 $\pm$ 0.00701	100 %	0.02764 $\pm$ 0.00684	100 %
mc_detumble_ais	montecarlo	power_peak	1.5	0.3902 $\pm$ 0.169	100 %	0.402 $\pm$ 0.16	100 %
mc_fine_img	montecarlo	ape_los_p9973	0.01	0.01315 $\pm$ 0.00434	29 %	0.01456 $\pm$ 0.00486	25 %
mc_fine_img	montecarlo	ake_los_p9973	0.005	0.002894 $\pm$ 0.000605	100 %	0.002523 $\pm$ 0.000383	100 %
mc_fine_img	montecarlo	rate_stability_p9973	0.005	0.003738 $\pm$ 9.57e-05	100 %	0.003755 $\pm$ 0.00013	100 %
mc_fine_img	montecarlo	power_mean	2	1.558 $\pm$ 0.0021	100 %	1.558 $\pm$ 0.00149	100 %
mc_nadir_ais	montecarlo	ape_los_last_orbit	10	91.95 $\pm$ 46.4	0 %	88.47 $\pm$ 43.7	0 %
mc_nadir_ais	montecarlo	ake_los_last_orbit	5	2.086 $\pm$ 1.02	96 %	1.651 $\pm$ 0.729	100 %
mc_nadir_ais	montecarlo	power_mean	0.5	0.01165 $\pm$ 0.00288	100 %	0.01013 $\pm$ 0.00302	100 %
mc_slew_cmgy	montecarlo	settle_time_after_slew	20	13.14 $\pm$ 0.46	100 %	13.03 $\pm$ 0.785	100 %

campaign	type	metric	req	MATLAB mean ± std	MATLAB pass	engine mean ± std	engine pass
mc_slew_cm	montecarlo	ape_los_on_target_p9973	0.01	0.003931 ± 0.000635	100 %	0.003761 ± 0.000349	100 %
mc_slew_img	montecarlo	settle_time_after_slew	20	13.03 ± 0.959	100 %	13.84 ± 4.76	98 %
mc_slew_img	montecarlo	ape_los_on_target_p9973	0.01	0.00815 ± 0.00115	92 %	0.007886 ± 0.001	95 %

• MATLAB twin ♦ Rust engine



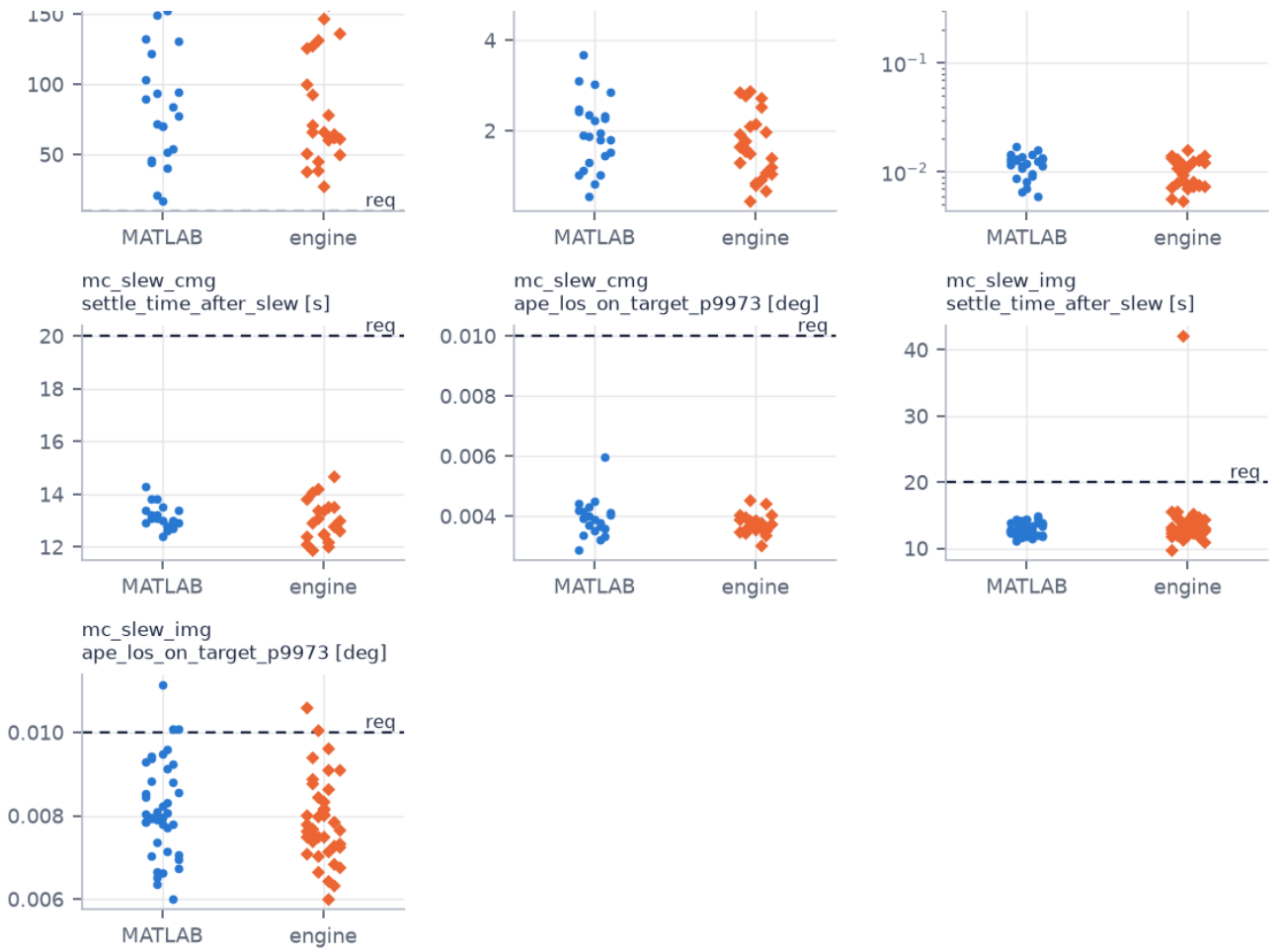


Figure 3. Every run of every campaign, requirement metrics: MATLAB twin (circles) and Rust engine (diamonds), dashed line = requirement.

10. Engine vs MATLAB twin parity

Single runs, engine vs MATLAB twin, same scenario, case and product: **94 of 107** requirement verdicts agree. The truth environment (orbit, Sun, Moon, shadow, density, field) of the engine's POP port is identical to the twin's on every filed sample of every scenario; what differs is the random stream (unit errors, initial tumble), so single-run metrics scatter inside the distributions of section 6. The differences traced to their cause are listed in results/ENGINE_PARITY.md.

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
agile_slew_cmgy	6001	0	0	0	
agile_slew_fmr_rcs	6001	0	0	0	
agile_slew_img	6001	0	0	0	
agile_slew_rw_rcs	6001	0	0	0	
agile_slew_vscmg	6001	0	0	0	
detumble_ais	17221	0	0	0	
detumble_ais_bangbang	17221	0	0	0	
detumble_ais_mag	17221	0	0	0	
detumble_img	17221	0	0	0	
fault_coil_ais	17221	0	0	0	
fault_gimbal_cmgy	57401	0	0	0	
fault_gyro_ais	17221	0	0	0	
fault_st_img	57401	0	0	0	
fault_wheel_img	57401	0	0	0	
fine_hold_cmgy	57401	0	0	0	
fine_hold_fmr	57401	0	0	0	
fine_hold_fmr_rcs	57401	0	0	0	
fine_hold_img	57401	0	0	0	
fine_hold_img_lqr	57401	0	0	0	
fine_hold_img_smc	57401	0	0	0	
fine_hold_rw_rcs	57401	0	0	0	
fine_hold_vscmg	57401	0	0	0	
mission_ais	28701	0	0	0	
mission_cmgy	17221	0	0	0	
mission_fmr	17221	0	0	0	
mission_fmr_rcs	17221	0	0	0	
mission_img	17221	0	0	0	
mission_rw_rcs	17221	0	0	0	
mission_vscmg	17221	0	0	0	
nadir_hold_ais	17221	0	0	0	

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
nadir_hold_ais_css	17221	0	0	0	
slew_cmgy	6001	0	0	0	
slew_fmry	6001	0	0	0	
slew_fmry_rcs	6001	0	0	0	
slew_img	6001	0	0	0	
slew_img_lqr	6001	0	0	0	
slew_img_smc	6001	0	0	0	
slew_rw_rcs	6001	0	0	0	
slew_vscmg	6001	0	0	0	
sun_spin_ais	17221	0	0	0	
agile_slew_cmgy	55	2.17	26x		
agile_slew_fmry_rcs	51	2.20	23x		
agile_slew_img	48	1.53	31x		
agile_slew_rw_rcs	55	1.52	36x		
agile_slew_vscmg	61	0.20	300x		
detumble_ais	496	0.97	511x		
detumble_ais_bangbang	848	0.74	1152x		
detumble_ais_mag	526	1.34	392x		
detumble_img	972	4.09	237x		
fault_coil_ais	533	0.59	908x		
fault_gimbal_cmgy	465	2.02	230x		
fault_gyro_ais	579	1.20	481x		
fault_st_img	389	1.74	223x		
fault_wheel_img	414	2.15	192x		
fine_hold_cmgy	468	2.58	182x		
fine_hold_fmry	659	1.77	371x		
fine_hold_fmry_rcs	523	2.28	230x		
fine_hold_img	390	1.38	282x		
fine_hold_img_lqr	388	1.91	203x		
fine_hold_img_smc	388	1.37	284x		
fine_hold_rw_rcs	394	1.72	229x		
fine_hold_vscmg	476	2.23	213x		
mission_ais	906	0.78	1158x		
mission_cmgy	1421	5.67	251x		
mission_fmry	1622	4.49	361x		
mission_fmry_rcs	1619	4.82	336x		
mission_img	1235	4.01	308x		
mission_rw_rcs	1263	4.95	255x		

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
mission_vscmg	1461	5.92	247x		
nadir_hold_ais	569	0.64	892x		
nadir_hold_ais_css	550	0.66	836x		
slew_cmg	54	0.54	100x		
slew_fmr	79	0.22	358x		
slew_fmr_rcs	56	0.22	249x		
slew_img	46	0.62	74x		
slew_img_lqr	46	0.59	77x		
slew_img_smc	45	0.24	192x		
slew_rw_rcs	46	0.18	260x		
slew_vscmg	56	0.33	168x		
sun_spin_ais	501	0.62	809x		

Verdicts that differ (all traced; see ENGINE_PARITY.md):

scenario	metric	MATLAB	engine	req	MATLAB	engine
fine_hold_fmr	power_mean	4.348	0.08406	2	FAIL	pass
fine_hold_fmr_rcs	power_mean	4.367	0.08406	2	FAIL	pass
fine_hold_img	ape_los_p9973	0.009166	0.01049	0.01	pass	FAIL
fine_hold_rw_rcs	ape_los_p9973	0.008375	0.03388	0.01	pass	FAIL
fine_hold_rw_rcs	rate_stability_p9973	0.003604	0.005641	0.005	pass	FAIL
mission_img	ape_los_last_half_orbit_p9973	0.007627	0.01066	0.01	pass	FAIL
mission_vscmg	ape_los_last_half_orbit_p9973	0.01111	0.009595	0.01	FAIL	pass
nadir_hold_ais	ape_los_last_orbit	7.248	12.64	10	pass	FAIL
nadir_hold_ais	ake_los_last_orbit	3.864	5.491	5	pass	FAIL
slew_img_smc	settle_time_after_slew	3.2	—	20	pass	FAIL
sun_spin_ais	sun_spin_entry	40.44	142.4	95	pass	FAIL
sun_spin_ais	sun_angle_last_orbit	6.503	93.84	20	pass	FAIL
sun_spin_ais	spin_rate_error_last_orbit	0.04568	1.116	0.5	pass	FAIL

11. Soft OILS

Every scenario flown twice on the engine: SILS (C flight software in-process) and soft OILS (the flight software compiled for the OBC, arm-none-eabi-gcc -mcpu=cortex-m4 -mfloat-abi=hard, running as firmware on QEMU mps2-an386 behind adcs-link/1). In soft OILS each command reaches the actuators only after the OBC's sensor reads, its execution and its CAN frames; the execution is the step's exact instruction count (QEMU plugin) $\times$ CPI 1.25 / 168 MHz. **105 of 107** requirement verdicts agree between SILS and soft OILS over 40 scenarios; runs are deterministic.

scenario	period [ms]	instructions/step mean / max	exec max [ms]	latency mean / max [ms]	C
agile_slew_cmgy	100	3.5e+05 / 8.94e+05	6.65	4.2 / 8.24	
agile_slew_fmr_rcs	100	3.01e+05 / 8.45e+05	6.29	3.31 / 7.36	
agile_slew_img	100	2.94e+05 / 8.44e+05	6.28	3.14 / 7.22	
agile_slew_rw_rcs	100	3.04e+05 / 8.51e+05	6.33	3.34 / 7.4	
agile_slew_vscmg	100	3.68e+05 / 9.17e+05	6.82	4.33 / 8.41	
detumble_ais	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
detumble_ais_bangbang	200	1.18e+05 / 4.64e+05	3.45	1.43 / 4	
detumble_ais_mag	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
detumble_img	100	8.12e+04 / 4.89e+05	3.64	1.55 / 4.58	
fault_coil_ais	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
fault_gimbal_cmgy	100	3.62e+05 / 8.98e+05	6.68	4.29 / 8.27	
fault_gyro_ais	200	1.93e+05 / 6.37e+05	4.74	1.99 / 5.3	
fault_st_img	100	2.54e+05 / 8.42e+05	6.26	2.84 / 7.21	
fault_wheel_img	100	2.9e+05 / 8.44e+05	6.28	3.1 / 7.22	
fine_hold_cmgy	100	3.62e+05 / 8.97e+05	6.68	4.29 / 8.27	
fine_hold_fmr	100	3.1e+05 / 8.45e+05	6.29	3.25 / 7.23	
fine_hold_fmr_rcs	100	3.13e+05 / 8.48e+05	6.31	3.4 / 7.39	
fine_hold_img	100	3.1e+05 / 8.45e+05	6.29	3.25 / 7.23	
fine_hold_img_lqr	100	3.1e+05 / 8.46e+05	6.29	3.25 / 7.24	
fine_hold_img_smc	100	3.13e+05 / 8.48e+05	6.31	3.27 / 7.26	
fine_hold_rw_rcs	100	3.15e+05 / 8.52e+05	6.34	3.42 / 7.42	
fine_hold_vscmg	100	3.83e+05 / 9.19e+05	6.84	4.44 / 8.43	
mission_ais	200	1.86e+05 / 7.58e+05	5.64	1.94 / 6.2	
mission_cmgy	100	3.05e+05 / 8.97e+05	6.67	3.87 / 8.27	
mission_fmr	100	2.61e+05 / 8.46e+05	6.3	2.89 / 7.24	
mission_fmr_rcs	100	2.64e+05 / 8.49e+05	6.32	3.04 / 7.39	
mission_img	100	2.68e+05 / 8.45e+05	6.29	2.94 / 7.23	
mission_rw_rcs	100	2.74e+05 / 8.55e+05	6.36	3.12 / 7.43	
mission_vscmg	100	3.23e+05 / 9.18e+05	6.83	3.99 / 8.42	

scenario	period [ms]	instructions/step mean / max	exec max [ms]	latency mean / max [ms]	C
nadir_hold_ais	200	1.93e+05 / 6.37e+05	4.74	1.99 / 5.3	
nadir_hold_ais_css	200	1.93e+05 / 6.38e+05	4.74	1.99 / 5.3	
slew_cmng	100	3.69e+05 / 8.94e+05	6.65	4.34 / 8.24	
slew_fmr	100	3.17e+05 / 8.41e+05	6.26	3.3 / 7.2	
slew_fmr_rcs	100	3.19e+05 / 8.44e+05	6.28	3.45 / 7.36	
slew_img	100	3.16e+05 / 8.42e+05	6.26	3.3 / 7.21	
slew_img_lqr	100	3.17e+05 / 8.42e+05	6.27	3.3 / 7.21	
slew_img_smc	100	3.19e+05 / 8.45e+05	6.29	3.32 / 7.23	
slew_rw_rcs	100	3.21e+05 / 8.47e+05	6.31	3.46 / 7.38	
slew_vscmg	100	3.89e+05 / 9.15e+05	6.81	4.49 / 8.4	
sun_spin_ais	200	1.21e+05 / 4.65e+05	3.46	1.45 / 4.01	

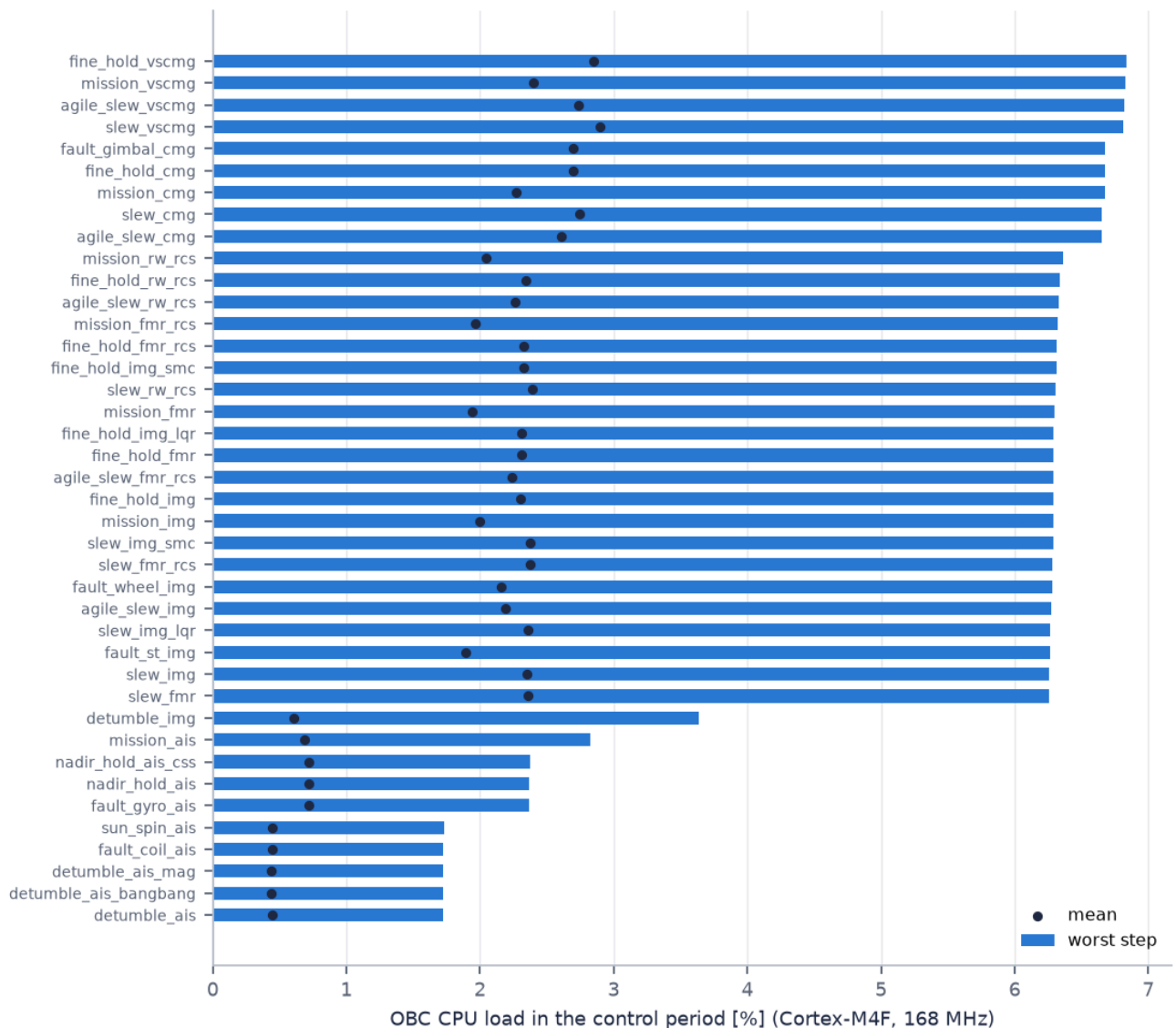


Figure 4. OBC load per scenario: worst step (bar) and mean (dot).

Verdicts that change between SILS and soft OILS (the command latency's effect):

scenario	metric	req	SILS	soft OILS	SILS	soft OILS
agile_slew_vscmg	ape_los_on_target_p9973	0.01	0.01008	0.009977	FAIL	pass
sun_spin_ais	sun_spin_entry	95	142.4	59.66	FAIL	pass

12. Virtual OBC bit-identity

Zero-latency lockstep (no timing model): the flight software as a host process and as QEMU Cortex-M4F firmware, C and Rust, reproduces the in-process trajectory bit for bit: **35 of 35** runs bit-identical.

scenario	reference	virtual OBC	result	wall [s]
detumble_ais	c (in-process)	obc-posix	bit-identical	0.3
detumble_ais	rust (in-process)	obc-posix-rs	bit-identical	0.2
detumble_ais	c (in-process)	qemu	bit-identical	18.2
detumble_ais	rust (in-process)	qemu-rs	bit-identical	17.4
detumble_ais	c (in-process)	rust	bit-identical	0.2
mission_ais	c (in-process)	obc-posix	bit-identical	0.2
mission_ais	rust (in-process)	obc-posix-rs	bit-identical	0.2
mission_ais	c (in-process)	qemu	bit-identical	14.2
mission_ais	rust (in-process)	qemu-rs	bit-identical	15.4
mission_ais	c (in-process)	rust	bit-identical	0.1
fine_hold_img	c (in-process)	obc-posix	bit-identical	0.6
fine_hold_img	rust (in-process)	obc-posix-rs	bit-identical	0.5
fine_hold_img	c (in-process)	qemu	bit-identical	55.9
fine_hold_img	rust (in-process)	qemu-rs	bit-identical	61.1
fine_hold_img	c (in-process)	rust	bit-identical	0.3
slew_img	c (in-process)	obc-posix	bit-identical	0.6
slew_img	rust (in-process)	obc-posix-rs	bit-identical	0.5
slew_img	c (in-process)	qemu	bit-identical	62.8
slew_img	rust (in-process)	qemu-rs	bit-identical	63.3
slew_img	c (in-process)	rust	bit-identical	0.3
fine_hold_cmng	c (in-process)	obc-posix	bit-identical	0.6
fine_hold_cmng	rust (in-process)	obc-posix-rs	bit-identical	0.6
fine_hold_cmng	c (in-process)	qemu	bit-identical	64.2
fine_hold_cmng	rust (in-process)	qemu-rs	bit-identical	68.9
fine_hold_cmng	c (in-process)	rust	bit-identical	0.3
fine_hold_fmr_rcs	c (in-process)	obc-posix	bit-identical	0.4
fine_hold_fmr_rcs	rust (in-process)	obc-posix-rs	bit-identical	0.4
fine_hold_fmr_rcs	c (in-process)	qemu	bit-identical	57.5
fine_hold_fmr_rcs	rust (in-process)	qemu-rs	bit-identical	57.1
fine_hold_fmr_rcs	c (in-process)	rust	bit-identical	0.5

scenario	reference	virtual OBC	result	wall [s]
sun_spin_ais	c (in-process)	obc-posix	bit-identical	0.2
sun_spin_ais	rust (in-process)	obc-posix-rs	bit-identical	0.2
sun_spin_ais	c (in-process)	qemu	bit-identical	15.3
sun_spin_ais	rust (in-process)	qemu-rs	bit-identical	13.6
sun_spin_ais	c (in-process)	rust	bit-identical	0.1

13. Node-by-node verification

231 of 231 checks pass over 20 nodes. Each check recomputes a node's decision from what it stored and the rules in the node registry (the catalogue pick from the datasheets, the selection from the family scores, the budgets from their units, each feasibility from its failures), rather than reading the node's own verdict. Every check is listed in `results/NODE_VERIFICATION.md`.

node	checks passed	verdict	failures
nodes	1/1	pass	—
case	2/2	pass	—
catalogue	51/51	pass	—
demand	6/6	pass	—
size_mtg	4/4	pass	—
size_fmr	8/8	pass	—
size_rcs	2/2	pass	—
select_rotor	12/12	pass	—
size_sensors	18/18	pass	—
budget	18/18	pass	—
matrix	2/2	pass	—
assess	4/4	pass	—
tune	14/14	pass	—
converge	8/8	pass	—
select	24/24	pass	—
dispatch	4/4	pass	—
certify	18/18	pass	—
mc	4/4	pass	—
family_missions	20/20	pass	—
soft_oils	8/8	pass	—
report	3/3	pass	—

14. Open items

- Bang-bang B-dot (fixed): pure sign switching limit-cycled around the 0.5 deg/s exit rate in 4 of 12 seeds with or without OBC latency; a boundary layer (4 x the B-dot gain inside it) detumbles 12 of 12 at 0, 4 and 8 ms. C, Rust, MATLAB and pseudocode updated.
- Rate stability (imaging): relaxed by the owner from 0.001 to 0.005 deg/s (18 arcsec/s, ~0.5 m smear in a 10 ms exposure at 550 km). At 0.001 deg/s it was set by gyro noise for every actuator (0.0025-0.004 deg/s with TRN-GYRO-P1) and needed a fibre-optic-class gyro (0.6 kg); at 0.005 deg/s the catalogue gyro holds it. The ADCS budget is 1.6 kg / 1.0 L (1U of the 3U; ceiling 1.7 kg).
- Fluid loop fine pointing: limited by the loop's flow sensor, not the pump. The converged imaging design carries a 0.125 mm/s (1 sigma) flow sensor; this is a requirement on the in-house sensor.

One star-tracker head instead of two breaks nadir pointing, so both stay.

- Engine vs MATLAB on fine_hold_rw_rcs rate stability: 0.0056 against 0.0036 deg/s (ratio 1.56, as before the relaxation); the new 0.005 deg/s line now falls between them, so this verdict disagrees. The gap is the wheel-plus-RCS model, not the requirement.
- Sun spin: 17 of 24 seeds pass with and without the OBC; failures are Sun-spin entry with the Sun near the XY plane tripping the 1 deg/s exit guard during the L2 precession transient -- a tuning trade (fsw.sun_spin_perp_out_dps, fsw.sun_spin_dwell_out_s).
- Coils only (mtq): sufficient for detumble and for Sun acquisition by Sun spin (the power face comes towards the Sun, though not within the cases' 95 min / 20 deg line). Nadir: with the hand-over (despin, then capture) and the capture test, see section 7 for the tuned laws and the mission figure. Three-axis Sun referencing stays out of reach with coils only (gravity gradient along B).
- Navigation and guidance audit (docs/NAV_GUIDANCE_AUDIT.md), corrected in C, Rust and the spec: the magnetometer update never ran while the coils actuated; the estimate was never dropped after the spin states; no innovation gating and fixed Sun/field sigmas; two-body-only onboard orbit; GNSS taken as J2000 and the field frame without precession (0.37 deg); in nadir the power face pointed away from the Sun on both cases (now yaw-flipped, 90 deg - $|\beta|$); nadir gravity-gradient feed-forward for ais_img_3u.
- Power face in coils-only nadir: Celani 2026 leaves the roll about the payload axis free; a weak roll PD (3n, zeta 1, certified $|\mu|$ 0.913) now turns the face towards the Sun (86 deg mean on ais_3u, was shaded) at the cost of pointing margin (6.6 deg nominal), since the coils' along-B shortfall lands on roll. Open from the audit: gyro scale factor aliases into bias after the despin; IGRF-13 is extrapolated for 2027 (load IGRF-14); nadir is geocentric, not geodetic; the MATLAB twin keeps J2000 GNSS, GMST-only frames and no gating.
- Bought against designed (both cases, same budget of 1.6 kg and 1.0 L): on ais_img_3u three CubeSpace CW0017 wheels are feasible at 1.0 kg against our fluid loop's 1.6 kg, so the lightest configuration overall is a benchmark; the fluid loop is selected because the selection is among our solutions, and its 0.6 kg is the price of not buying wheels. On ais_3u every rotor fails the 0.5 W orbit-average power (three CW0017 draw 0.9 W steady) and the fluid loop, feasible at 1.45 kg, is the only one that passes. The Tensor Tech CMG cluster fails power in both cases on its whole-ADCS datasheet figure (4 W upper bound).
- Catalogue data: the vendor sites were not reachable from the build environment. CubeSpace Gen2, Rocket Lab RW-0.01, RW3-0.06, RW3-1.0 and Tensor Tech ADCS100/400 numbers were read from the full datasheets (distributor mirror); AAC Clyde Space RW222/RW400, Rocket Lab RW-0.03 and Tensor Tech CMG-10m from search excerpts. AAC Clyde Space models lack mass or power on what could be read, so they are listed but not selectable; confirm with the vendors before a buy decision.
- Soft OILS models the OBC's CPU and buses; real OILS still needs the board target (fsw/targets/<board>/main.c: UART/Ethernet, clock, linker script, modelled on fsw/targets/qemu-mps2) and --realtime on adcs-link/1.